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An introduction to second language speech research

If you are reading this book, you are likely a second/third language learner curious about her or his pronunciation, a linguist, psychologist, or other researcher interested in issues of bi/multilingualism and learning with a particular focus on speech, or both. Regardless of your interests, there are likely to be various aspects of your own speech or that of non-native speakers that have struck you and that you wish to understand better. In this book, we will investigate what is commonly referred to as **foreign accent**, the various differences between the speech perception and production of non-native and (sometimes idealized) native speakers. Our goal is to provide you with an understanding of the principles and phenomena of second language (L2) speech perception and production, theories that have been developed to model these, and the experimental methodologies used to investigate both **segmental** (i.e., consonants and vowels) and **prosodic** phenomena (e.g., lexical stress or tone, rhythm, intonation, and fluency). Our hope is that, once you have finished reading this book, you will either have answers to your questions from previous research or, perhaps more importantly, the ability to conduct your own studies.

In this first chapter, we lay the foundation for the discussion in the entire book, focusing on the major theoretical and empirical questions that guide L2 research, particularly as concerns the acquisition of phonetics and phonology (§1.3). Each theme will be introduced via a series of research questions to be explored throughout the book followed by the presentation of an illustrative study. However, before discussing these central themes of L2 speech research, we first examine the basic structure of speech (§1.1) and a number of concepts relevant to any study of L2 acquisition (§1.2).

1.1 The structure of speech

Human speech is the focus of two major branches of linguistics, namely **phonetics** and **phonology**. Traditionally, these two subdisciplines have been distinguished in terms of their orientation to the study of spoken language. Phonetics investigates the physical aspects of sound. These include its production (articulatory phonetics), transmission (acoustic phonetics), and perception (perceptual phonetics). Phonology, in contrast, focuses more on the abstract organization of sound systems such as the ways in which sounds may be contrastive and convey meaning (e.g., minimal pairs such as English *pay* /pe/ and *bay* /be/; French *frais* /fʁɛ/ 'fresh' and *vrai* /vʁɛ/ 'true'; and Spanish *gata* /gata/ 'cat' and *rata* /rata/ 'rat'). Phonology also interacts with other linguistic domains including morphology (word structure), syntax (sentence/utterance structure), semantics (meaning), and pragmatics (socially and context-appropriate language use) observed with phenomena including stress and intonation. In the framework adopted in this book, namely that of **experimental phonology**, phonetics and phonology are seen to a great extent as intricately interwoven.

When analyzing both speech perception and production, phoneticians and phonologists normally distinguish between two levels of organization, namely segments and prosody. The study of segments focuses not only on the individual realization of consonants and vowels but also on their production in sequences, that is on their **coarticulation**. In contrast, prosody refers to all phenomena that serve to group individual sounds into larger units including syllables, metrical feet relevant for the assignment of stress, and even larger units including phonological phrases and utterances necessary for the organization of phenomena such as intonation; prosody also looks at fluency phenomena including rhythm and timing. In Part III of this book, each chapter will begin with a thorough discussion of the phonetics and phonology of vowels (Chapter 4), consonants (Chapters 5, 6), sequences (Chapter 7), or prosody (Chapter 8) necessary for understanding and undertaking L2 speech research. At the end of this chapter, you will also find some useful readings if you wish to explore further these two subdisciplines of linguistics.

1.2 Core concepts and terminology in second language acquisition

Within the larger field of bilingualism and multilingualism, **second language (L2) acquisition** refers to the acquisition of a language following the (mainly complete) acquisition of one's first language (L1). In those cases involving child learners where acquisition begins before the age of seven to ten years and results in very proficient mastery of the target language, researchers often refer to **child** or **early L2 acquisition**. Unless otherwise highlighted, the research discussed in this book focuses on the former, more common type of L2 acquisition.

In more recent years, researchers have begun to distinguish clearly between L2 acquisition and **third language (L3) acquisition**. This latter type of learning involves the acquisition of a non-native language subsequent to that of another non-L1 (see De Angelis (2007) for a general overview and Wrembel, Gut & Mehlhom (2010) for a collection of recent studies on L3 phonology in particular). With the exception of this more recent research, researchers have most often not distinguished between L2 and L3 acquisition. Accordingly, in this book, with the exception of those instances where we discuss studies that seek to investigate L3 acquisition in particular including the effect of a previously acquired L2 on learners' perception and production, we will adopt the more common approach of referring to all such learning as L2 acquisition.

In L2 acquisition, learners begin with the knowledge of their L1 as well as of any other previously (partially) acquired non-native languages, and then move gradually towards the language being acquired, or **target language (TL)**. Acquisition is driven by the ambient spoken and written language, that is, the **input** to which learners are exposed. Over the course of acquisition, learners' perceptual and productive knowledge of the target language will be represented cognitively as a series of **interlanguage (IL) grammars**. The term *interlanguage* (e.g., Corder, 1967; Selinker, 1972; Adjemian, 1976) refers to the fact that, once acquisition has begun, learners' L2 perceptual and production knowledge normally falls somewhere in between that of their L1, due to **cross-linguistic influence (CLI)** of this dominant language, and the target language. Often, in spite of the variability observed between learners of a given target language who share the same L1 as well as with an individual learner at a given point in time, it is possible for researchers to distinguish **developmental sequences**. These are universally observed patterns of acquisition that involve a series of stages characterized by patterns and processes common to all learners regardless of their L1. The transition from one stage of development to the next is gradual, with the rate of change varying from learner to learner based on a number of variables including the quantity and quality of input and a learner's overall aptitude for L2 acquisition as determined by factors including phonological memory and motivation. Furthermore, features of more than one of the stages identified by researchers may be observed in learners' perception or production at a given point in time. If a learning plateau is observed during which acquisition appears to have stopped, even temporarily, one speaks of **fossilization**. At the point at which it appears that acquisition is complete or will advance no further, the IL grammar may be referred to as an **end-state** grammar and the characteristics of this grammar are referred to as a learner's **ultimate attainment**, which may be more or, more commonly, less native-like. In the domain of phonetics and phonology, L2 end-state grammars typically, but not necessarily, differ from those of native speakers. As we discussed at the beginning of this chapter, this is referred to as foreign accent.

In the next section, we will investigate the research questions that specifically target each of these concepts and processes related to the L2 acquisition of speech.

Each of these will be exemplified with some previous research including a series of illustrative studies.

1.3 L2 speech research questions

All research is driven by the desire to answer one or more questions concerning a given phenomenon; work on L2 speech is no exception. In the following sections, we discuss many of the recurrent theoretical themes found in experimental work on L2 speech perception and production. Answers to these questions, at times partial, will be offered in the chapters that follow.

1.3.1 The role of input

Theories of language acquisition assign a primary role to the linguistic input based on which learners construct both perceptual and production grammars. As mentioned earlier, in the case of speech learning by literate learners, input consists of both aural and written language.

The first question that one can identify in previous research is *How does input trigger speech learning?* This question may be of a particular interest in the context of L2 acquisition given that, even when faced with large quantities of native input, learners may fail to acquire native-like perceptual and/or productive competence. How one answers this question depends much on the particular theoretical framework adopted. Two broad theoretical frameworks can be identified. The first is that of researchers who assume that language learning, including L2 speech acquisition, is primarily associative, with learners establishing connections between form and meaning. Such an approach is variably referred to as **functionalist** or **associative-connectionist** (see, e.g., Ellis (2007) for a general discussion of this approach to L2 acquisition). On this approach, input drives acquisition with learners forming generalizations based on the language to which they are exposed. Such generalizations principally involve associations between a form in a given context and its meaning. For example, in the case of the acquisition of the phonemic inventory of a given target language, once learners have begun to be able to segment the speech signal into words, they begin to identify minimal pairs from which they can deduce which phones are contrastive and thus phonemic. The second broad theoretical framework includes researchers who adopt **nativist** models. Such models propose that humans are endowed with a language faculty, which includes innate knowledge concerning the universal set of possible linguistic categories, structures, and principles of their organization. In the domain of phonology, this would include the possible features that may be used to establish phonemic contrasts, minimal and maximal syllable templates, and the basic units and principles used to construct the representations of stress systems. The most commonly encountered work within this framework is conducted within Chomsky's Universal Grammar (see White (2003) for detailed discussion of this framework for the L2 acquisition of morphosyntax and Archibald (1998) for its use in modelling the acquisition of L2 phonology). Within nativist models, input serves to trigger language-specific settings of parameters governing the wellformedness of linguistic representations or, in constraint-based frameworks, to allow for the ranking of constraints that evaluate wellformedness and complexity. The framework within which one undertakes L2 speech research is a function of a researcher's larger understanding of the fundamental nature of human language and, one must admit, the community of researchers within which one was trained and currently works. The overwhelming majority of previous L2 speech research has adopted a functionalist approach.

A second principal question related to the role of input is *What are the consequences of differences in input quantity, quality, and modality for L2 speech learning*? Given the primary importance of input, these three variables each have an enormous potential effect on the type of cross-linguistic influence observed, patterns of development, and learners' ultimate attainment including the degree of intra- and interspeaker variability. The input to which a given group of L2 learners has access may differ greatly in quantity and quality as well as **modality** (i.e., auditory, visual, or written) vis-à-vis the input available to children or other L2 learners. In terms of quantity, children are exposed to vast amounts of auditory native speaker input from birth. This large quantity of ambient language allows child learners to perceive and, with sufficient practice, produce the full range of segmental and prosodic phenomena of the target language being acquired. In contrast, the input available to many L2 learners is often impoverished both in terms of quantity and quality (see Moyer (2009) for an overview). For example, when a target language is encountered principally in a formal learning environment such as the classroom and is not the majority language of the surrounding speech community, learners may be limited to a few hours of input a week from a single individual who may or may not be a native speaker. In a classroom setting, the speech of one's fellow learners also constitutes input to learning, input which is clearly deficient, at least in terms of quality. The consequence of such a situation is that learners may come to acquire perception and production grammars that accurately reflect the input to which they were exposed, although such input may not reflect the speech of native speakers. As mentioned above, differences in input also involve modality. Whereas child learners do not have access to written language until the onset of the acquisition of literacy, most adult learners are literate, at least those who have served as participants in the vast majority of the studies reported in published research (see Young-Scholten, 1995, p. 114 for more discussion). We will return to the question of the effect of literacy, particularly that of written input, on L2 speech learning in §1.3.6. Within research on the effects of differences in quality and quantity of input, a particular sub-domain studies the potentially facilitative effect of training in the laboratory (e.g., Iverson, Hazan & Bannister, 2005; Motohashi-Saigo & Hardison, 2009) and instruction in the classroom (e.g. Macdonald, Yule & Powers, 1994; Saito & Lyster, 2012). Such work investigates the degree to which various techniques and paradigms including high variability training, acoustic signal manipulation, and form-focused instruction allow L2 learners to more effectively analyze the input when acquiring both segmental and prosodic aspects of the TL.

Illustrative study 1.1. Freed, Segalowitz & Dewey (2004)

Overview: This study investigated differences in gains in oral performance and fluency between students in three learning contexts differing in input quantity and quality: traditional classroom study in the home country (AH – at home), intensive traditional classroom study accompanied by a variety of out-of-class activities in the target language (IM – immersion), and a study abroad (SA) experience combining formal instruction and possibilities for regular interaction with the native language speech community.

Languages: French (TL); English (L1)

Dependent variables: *Oral performance*: (1) Total words spoken; (2) Duration of speaking time; (3) Length (in words) of longest turn; *Oral fluency*: (1) Speech rate; (2) Hesitation-free speech runs; (3) Filler-free speech runs; (4) Fluent runs versus repetition-free speech runs; (6) Grammatical-repair-free speech runs.

Independent variable: *Learning context*: The three contexts differed in terms of the total number of hours of formal instruction on average (AH: 12 weeks, 36–48 hours; IM: 7 weeks, 123 hours; SA: 12 weeks, 197 hours) and opportunities for contact and use of French outside of class (AH: highly limited; IM: a variety of cultural and other activities throughout the period with students taking a 'language pledge' to communicate only in French including outside of class; SA: the programme took place in Paris with the majority of students participating in courses offered by their home institution)

Research questions: (1) Are there salient differences in the acquisition of oral fluency by students who have studied abroad, when compared to students whose learning takes place in IM programmes or the regular AH language classroom? (2) Do time-on-task factors (e.g., instructional time, out-of-class time spent interacting orally with native speakers or using the language within the literate domain) vary in each of these contexts? (3) To what extent are the measured differences in oral fluency associated with these time-on-task features? (p. 280)

Methodology

Participants: 28 adults (mean age = 21.3 years) studying in an American university
Tasks: (1) Oral interviews: 15–30 minutes in length, conducted both preceding and following the period of instruction; (2) Questionnaire: a language contact profile designed to determine language use (how much time was spent reading, listening, writing, and speaking and in which language) and interaction (with whom).
Data analysis: *Oral interviews*: For each participant, two one-minute samples from the pre- and post-course interviews were transcribed and verified by two sets of two individuals and then analyzed for the nine dependent variable temporal and hesitation phenomena. Fluency gain scores were calculated by subtracting pre-course from post-course scores.

Main findings

(1) Whereas the students in the three groups differed minimally on their pre-course measures, statistically significant gains were made only by the IM group and only for a subset of the measures of oral performance (total number of words, length of longest turn) and fluency (rate of speech). Moreover, for a composite fluidity variable (combined hesitation-free, filler-free, repetition-free, grammatical-repair-free, and fluent speech runs), the IM group made the most overall gains in fluency follow by the SA group; the AH group made no gains at all.
(2) The IM group reported the greatest number of hours of speaking, reading, writing, and listening to French.
(3) The only unambiguous, significant correlation discovered existed between the composite fluidity variable and time spent on writing.

1.3.2 Cross-linguistic influence

A widely observed phenomenon in any bi- or multilingual speaker is the interaction of the individual's various languages in both comprehension/perception and production. This phenomenon, alternatively referred to as cross-linguistic influence (CLI), **transfer**, or **interference**, occurs in L1 bilingual language acquisition (e.g., Paradis, 2001; Fabiano-Smith & Goldstein, 2010), L2 acquisition (e.g., Flege & MacKay, 2004; Lee, Guion & Harada, 2006), and adult balanced bilinguals (e.g., Fowler, Sramko, Ostry, Rowland & Hallé, 2008). In the case of L2 phonological acquisition in particular, CLI has been observed to influence individual segments (e.g., McAllister, Flege & Piske, 2002; Aoyama, Flege, Guion, Akahane-Yamada & Yamada, 2004) as well as sequences including coarticulation (e.g., Levy & Strange, 2008; Zsiga, 2003), and all aspects of prosody (e.g., Steele, 2002 for syllable structure; Matthews & Brown, 2004; Archibald, 1997; Dupoux, Pallier, Sebastián-Gallés & Mehler, 1997 for stress; Hallé, Chang & Best, 2004; Francis, Ciocca, Ma & Fenn, 2008 for tone; Aoyama & Guion, 2007; Nava & Zubizaretta, 2009 for rhythm; and Mennen, 2004; Swerts & Zerbian, 2010 for intonation).

One of the most obvious cases of CLI is a speaker's use of L1 categories or prosodic patterns in their L2 production. For example, French-speaking learners of English may realize the voiceless interdental fricative /θ/ of English words such as *think*, *athlete*, and *bath* as well as the voiced interdental fricative /ð/ of *this*, *although*, and *bathe* with their L1 /s/ and /z/ respectively (i.e., as [s]ink, [z]is) given the absence of the former consonants in French. Such substitutions are specific to a learner's particular L1 **variety** (i.e., dialect). Whereas the substitution pattern above holds true for European French speakers, native speakers of Quebec French substitute rather /t/ and /d/ (i.e., [t]ink, [d]is; e.g., Brannen, 2002; Picard, 2002; Trofimovich, Gatbonton & Segalowitz, 2007). Native dialect effects are also observed in perception. A variety of recent studies (Chládková & Podlipský, 2011; Escudero, Simon & Mitterer, 2012; Escudero & Williams, 2012; among others) have demonstrated that L2 listeners categorize TL vowels based not on the phonemic categories of their first language but rather the acoustic categories of their particular L1 variety. We will examine the effect of L1-based CLI on perception further in Chapter 2 when discussing theories of L2 speech perception.

At the prosodic level, the effect of L1-based CLI has been studied most often as concerns its consequences for syllable structure modification and misplacement of lexical stress. In terms of the former, learners may modify TL consonant sequences that exceed the complexity permitted in their L1, most often via vowel epenthesis (e.g., Broselow, Chen & Wang, 1998; Abrahamsson, 1999) or consonant deletion (e.g., Eckman, 1987; Hancin-Bhatt & Bhatt, 1997). For example, languages like English, French, and Spanish allow initial stop-liquid clusters (e.g., *play* /ple/, *drain* /dren/; *clouer* /klue/ 'nail (INF)', *privé* /pʁive/ 'private'; *globo* /globo/ 'balloon', *crema* /krema/ 'cream') not permitted in languages whose dominant syllable template is CV (consonant-vowel) or CVC (consonant-vowel-consonant) including Mandarin and Cantonese. L2 learners with such latter L1s may realize TL stop-liquid clusters via the insertion of a vowel (e.g., English *drain* /dren/ realized as [dəɹen]; Spanish *globo* /globo/ realized as [gəlobo]) or by deletion of one of the consonants, most commonly the liquid (e.g. English *play* /ple/ realized as [pe]; French *privé* /pʁive/ realized as [pive]). Epenthesis and deletion both serve to bring TL syllable structure into conformity with a learner's L1; we will discuss syllable structure modification in further detail in §7.4. In the case of lexical stress, CLI results in the application of L1 stress patterns to TL lexical items (e.g., Archibald, 1993a; 1997). In many cases, this results in the misplacement of stress. For example, a Spanish-speaking learner of English, whose L1 favours penultimate stress, might pronounce English *afford*, *concentrate*, *currently*, *innocent*, and *retrieve* stressing the second-to-last syllable in a non-target-like manner (i.e., *ásfford, concéntrate, curréntly, innócent*, and *rétrieve*; Archibald, 1993b). As was the case with segmentals, the effects of L1-based prosodic CLI are not restricted to production. In the case of syllable structure, a number of studies have demonstrated the existence of perceptual epenthesis (e.g., Dupoux, Kakehi, Hirose, Pallier & Mehler, 1999; Matthews & Brown, 2004; Kabak & Idsardi, 2007; Dupoux, Parlato, Frota, Hirose & Peperkamp, 2011) and perceptual deletion (Steele, 2009). Using minimal pair discrimination or word identification tasks, this research has shown that, when TL syllable structure exceeds the complexity permitted in the L1, learners' perceptual systems will accommodate the TL forms to their L1 resulting either in the perception of a vowel having no acoustic correlate or the deletion of one of the members of the cluster. In the case of lexical stress, a number of studies by Dupoux and colleagues (Dupoux, Pallier, Sebastián-Gallés & Mehler, 1997; Peperkamp & Dupoux, 2002; Dupoux, Sebastián-Gallés, Navarrete & Peperkamp, 2008; Dupoux, Peperkamp & Sebastián-Gallés, 2010; Peperkamp, Vendelin & Dupoux, 2010) have shown that listeners whose L1 has fixed, non-contrastive lexical stress are 'stress deaf' and have great difficulties distinguishing lexical items differing only in the location of word stress. We will discuss the specifics of perceptual modification and stress deafness in Chapters 7 and 8 respectively.

CLI phenomena do not involve only higher-level phonological phenomena such as syllable structure and lexical stress discussed immediately above. The interaction between a learner's L1 and L2/L3 perceptual and production grammars may also involve low-level phonetic phenomena including language-specific differences in the realization of segments and prosody. One frequently studied such difference is the ways in which languages realize the phonemic voicing contrast in stops. To illustrate, consider English, French, and Spanish. All three languages have a contrast between voiceless and voiced stops (e.g., English: /tu/ *two*, /du/ *do*; French: /po/ *peau* 'skin', /bo/ *beau* 'handsome'; Spanish: /tos/ *tos* 'cough', /dos/ *dos* 'two'). However, these languages differ in the way in which this contrast is implemented phonetically. One of the principal differences involves a parameter known as Voice Onset Time (VOT), which measures the temporal interval in milliseconds (ms) between the release of the stop closure and the onset of the glottal vibration (i.e., voicing) of the following vowel or approximant consonant. In English, phonemically voiced stops are realized with a relatively small VOT while voiceless stops involve a much longer VOT (depending upon place of articulation, approximately 0–20 ms versus 60–80 ms respectively; e.g., Lisker & Abramson, 1964). In contrast, in both French and Spanish, phonemically voiced stops have a negative VOT (i.e., laryngeal voicing begins before stop release; −140 to −110 ms) whereas voiceless stops have a short VOT very much like that observed with English phonemically voiced stops. When lower proficiency English-speaking learners produce French or Spanish stops, these consonants are typically realized with English VOT values or values closer to those of English than the target VOT. Together, CLI at both the phonological and phonetic level, at least in those cases where it results in non-target-like production, contributes greatly to perceived foreign accent. We will discuss all of these patterns in greater detail in §5.2.1.

Broadly speaking, previous work on CLI has focused on two main research questions. The first of these is *How do previously acquired languages, both the L1 and other L2s, affect learners' perception and production?* In this section, we have already discussed some particular consequences of CLI for both perception and production. As concerns perception in general, it has been demonstrated time and time again that a hearer's L1 serves as a type of filter that groups TL sounds with reference to existing L1 (and, potentially, L2) categories based on the degree of perceived similarity. The formalization of L1-based CLI on perceptual categorization, including whether categorization is based on primarily phonetic (acoustic or articulatory) or phonological categories, has led to the proposal of a number of theories of L2 speech perception, which will be discussed in §2.2. No parallel theories of L2 production exist that formalize CLI in production. However, researchers have drawn on both phonetic theories including Articulatory Settings theory (Wenk, 1979, 1983; Wilson, 2006) and phonological frameworks such as Optimality theory (e.g., Hancin-Bhatt, 2000; Escudero & Boersma, 2004; Broselow & Xu, 2004) to this end. The former will be discussed in detail in §2.3.1.

The second question, *What factors best predict the nature of CLI?*, has received relatively less attention. This question is no less interesting, as it is not uncommon for L2 learners with the same L1 to demonstrate different patterns of L1 influence or for a given learner's perception or production to be variably influenced by his or her other languages. Within the growing field of L3 acquisition, determining which factors shape patterns of L1 influence has been a major area of research. Such work has revealed that various factors including interlinguistic typological similarity, language dominance and proficiency, recency of use, and the degree of exposure to the L2 and L3 (i.e., input quality and quantity) are among the best predictors of patterns of CLI (e.g., Gallardo del Puerto, 2007; Llama, Cardoso & Collins, 2010; Escudero, Broersma & Simon, 2013; Escudero & Williams, 2012).

Illustrative study 1.2. Escudero & Williams (2012)

Overview: This study investigated the effect of L1-based CLI on L2 perception. This was done by testing the effects of acoustic differences between two Spanish varieties in the realization of the five-vowel inventory /a,e,o,i,u/ on the L2 perception of Dutch vowels, controlling for the effect of L2 proficiency.

Languages: Dutch (TL); Iberian Spanish (IS), Peruvian Spanish (PS) (L1s)

Dependent variable: Perceptual accuracy – participants were tested on their perception of 12 Dutch vowels /ɑ,a,ɛ,e,ø,ɪ,i,ɔ,u,o,ʏ,y/ as measured by perceptual accuracy scores (percent correct categorization or identification). Of the five vowels shared with Spanish, there are a series of between-variety differences that have the following two important consequences for the L2 perception of the Dutch vowels in terms of L1-dialect influence: (1) For the Dutch pair /i-ɪ/: Whereas IS /i/ is acoustically closer to Dutch /ɪ/, PS /i/ is midway between the two TL vowels; (2) For the Dutch pair /a-ɑ/: The IS realization of /a-o/ is more similar to the Dutch pair than the realizations of PS.

Independent variables: (1) *L1 variety* [Iberian, Peruvian Spanish]; (2) *TL proficiency* [beginner, intermediate, advanced]

Hypotheses: (1) *L1 dialect effects*: The acoustic differences between IS and PS Spanish will result in differences in perceptual accuracy with the Dutch vowels. Specifically, for /i-ɪ/, IS hearers should be more accurate given that their L1 /i/ maps closely to Dutch /ɪ/, whereas PS /i/ can map to either target vowel. In the case of Dutch /a-ɑ/, the greater acoustic similarity of Spanish /a-o/ in IS than PS will result in greater perceptual accuracy in the IS group; (2) *TL proficiency*: If Dutch proficiency alone influences L2 vowel perception, the more proficient learner group should outperform their dialectal counterpart. Specifically, as the PS group was more proficient on average (see below), relatively overall higher perceptual accuracy was predicated for this group.

Methodology

Participants: 99 (48 IS, 51 PS) Spanish-speaking learners of Dutch living in the Netherlands at the time of testing.

Tasks: (1) *L2 proficiency*: Learners' Dutch proficiency – beginner, intermediate, or advanced – was determined using a general listening comprehension task; (2) *L2 perception* – Two-choice forced alternative (XAB) categorization: participants were presented auditorily with three vowels in succession and had to indicate whether the first vowel (X) was more like the second (A) or third (B); (3) *L2 perception* – 12-alternative forced choice identification: participants were presented a single vowel auditorily then required to indicate which vowel they had heard using one of 12 orthographic responses.

Stimuli: The 12 Dutch vowels /ɑ,a,ɛ,e,ø,ɪ,i,ɔ,u,o,ʏ,y/ were produced by 20 native speakers (10 male, 10 female) in the context [s_s]. In Task 2 (categorization), participants were tested only on the five pairs /a-ɑ/, /i-ɪ/, /y-ʏ/, /i-y/, and /ɪ-ʏ/. Whereas the first vowel (X) was from the naturally produced set, the second (A) and third (B) vowels were synthesized based on the mean F1 (height) and F2 (backness) values of the 10 male speakers. In Task 3 (identification), participants were tested on all 240 (12 vowels × 20 speakers) natural stimuli.

Data analysis: The percentage of accurate responses was calculated then statistical between-group comparisons were made.

Main findings

Proficiency: Both groups included learners at all three levels of proficiency (IS: 27 beginner, 6 intermediate, 15 advanced; PS: 12 beginner, 18 intermediate, 21 advanced). As a group, the PS learners were significantly more proficient (average Level 5 or 6 compared to Level 1 with the IS learners).

Hypothesis 1: *Effect of dialect*: (1) Task 2: significant differences were found for two of the five pairs (/a-ɑ/, /i-ɪ/) with the PS group being less accurate on both (55% versus 61% and 63% versus 68% respectively; (2) Task 3: while no significant main effect was found for dialect, a significant secondary effect for vowel-by-dialect interaction existed with the two groups differing in identification accuracy with /ɔ/; the PS group was once again less accurate (56% versus 75%).

Hypothesis 2: *Effect of proficiency*: (1) Task 2: No significant correlations were found between proficiency and categorization accuracy. In contrast, a significant correlation was found between dialect scores for target Dutch pair /a-ɑ/; (2) Task 3: While two significant correlations were found between L2 Dutch proficiency and identification accuracy, six significant correlations were found between dialect and identification accuracy. As the authors state, overall, native dialect effects better explain the observed differences than proficiency.

1.3.3 Developmental sequences and relative difficulty of acquisition

The study of linguistic development is critical to our understanding of acquisition more generally. While research that investigates learners' perception or production at a given moment in time allows for important insights into the results of acquisition, developmental studies, whether cross-sectional or longitudinal, provide a window onto the acquisitional process itself including onto changes in the relative influence of CLI and universals processes over time, and the ways in which learners analyze input as witnessed in developmental sequences.

Within L2 acquisition, the identification of such sequences – that is, series of sequential stages in which each stage is characterized by the presence of a number of linguistic forms and, sometimes, the absence of others – first focused on child L2 learners. In a series of oft-cited papers, Dulay & Burt (1973, 1974a, 1974b, 1975) identified an acquisitional order for English functional morphology (Stage 1: case; Stage 2: singular forms of copular *be*, auxiliary *be*, progressive *-ing*; Stage 3: irregular past forms, conditional *would*, possessive *-s*, plural *-(e)s*, third person singular *-s*; Stage 4: auxiliary *have*, past participle *-en*). Studies of adult L2 acquisition (Bailey, Madden & Krashen, 1974; Krashen, Butler, Birnbaum & Robertson, 1978) revealed the existence of similar sequences. Work on developmental sequences in L2 phonology has primarily focused on the production of segments including positionally-conditioned allophones (e.g., Trofimovich, Gatbonton &

Segalowitz, 2007; Munro & Derwing, 2008; Cardoso, 2011a) and syllable structure (e.g., Sato, 1987; Osburne, 1996; Carlisle, 1998; Abrahamsson, 2003; Hansen, 2004). Work on developmental sequences for prosody is strikingly absent (but see Tremblay, 2009) and, thus, an area ripe for research. Given the challenges of conducting longitudinal studies including ensuring the continued participation of adult learners, more of this work has been cross-sectional in nature with developmental stages inferred from the patterns that characterize learners of different proficiency levels.

The first two research questions that have been asked concerning developmental sequences are, *Are there universal patterns of L2 phonetic and phonological development? If so, what is their nature?* The answer to the first of these questions is 'yes'. As has been observed for morphosyntax, the acquisition of a given phonological structure involves stages of progressive development. For example, in the case of complex onset and coda consonant clusters, learners first acquire two-member sequences followed by those involving three consecutive consonants (e.g., Carlisle, 1998). It should be noted that, in spite of the existence of such sequences, it has been demonstrated that, with a given structure, learners may initially be more accurate, then less accurate, then more accurate once again (e.g., Carlisle, 1998, 2002; Abrahamsson, 1999, 2003). This developmental phenomenon is referred to as a **U-shaped learning pattern**. Alternatively, there may be an initial period of relatively rapid development followed by learning plateaux (e.g., Flege, 1988; Flege, Munro & Skelton, 1992; Munro & Derwing, 2008). While developmental sequences are universal by definition, there are nonetheless differences in the rate at which learners pass through one stage to the next. Researchers have thus asked the question *What factors shape interlearner differences?* Relevant factors, including quantity and quality of input, will be presented in §1.3.5 when discussing ultimate attainment.

Another related question is whether the progression through the various stages of developmental sequences occurs more readily with certain structures than others. Stated otherwise, acquisitionists have asked, *Is it possible to predict the relative acquisitional difficulty of structures?* For example, in the acquisition of the English vowel system by learners whose L1 only has five vowels (e.g., Spanish /a,e,i,o,u/), one could ask whether there are certain of the new English vowels – for example, /æ,ʌ/ as in *bat*, *butt* versus /ɛ,ɪ/ as in *bet*, *bit* – that are more difficult to acquire. If it is possible to determine which of two or more structures is most or least difficult, in order to make specific predictions, one then needs to ask, *What factors correlate with difficulty?* Answers to this latter question are varied, and focus on both perception and production. The first type of answer evokes the concept of **markedness** (see, e.g., Anderson, 1987; Carlisle 1991, 1997, 1998; Eckman & Iverson, 1994 for the use of markedness within L2 speech studies). While the relative markedness of two or more structures can be determined in a variety of ways including by considering cross-linguistic relative frequency and developmental orders in child acquisition, research on the role of markedness in L2 acquisition has most often defined it implicationally: for two related structures S_A and S_B, if the presence of S_B implies the presence of S_A cross-linguistically but the reverse implication does not hold, then S_B is said to be marked relative to S_A. For example, cross-linguistically, languages that allow the phonemically voiced stops /b,d,g/ syllable finally also allow the phonemically voiceless stops /p,t,k/ in this position. The reverse relationship does not hold. As such, cross-linguistically, in coda position, phonemically voiced stops are marked relative to their voiceless counterparts. Following Eckman (1977)'s Markedness Differential Hypothesis, structures that are absent from a learner's L1 and are typologically marked will be acquired with greater difficulty. Thus, in the case of the acquisition of coda stops by a learner whose L1 does not permit any stops, voiceless stops should be acquired more easily than their voiced counterparts. The second potential answer to the question of which factors predict relative difficulty comes from perception. As we will see in Chapter 2, one of the most widely cited theories of L2 perception, the Speech Learning Model (Flege, 1995; 2003), predicts that TL segments that are different but extremely similar to existing L1 categories (or, potentially, existing L2 categories in the case of L3 acquisition) will be the most difficult to acquire, as learners' perceptual systems will categorize the TL segments as existing categories and thus fail to create new, target-like ones. For example, if we consider the case of an English learner of French, one can predict the relative difficulty of acquiring the French mid-close front vowels /e,ø/. Rounded /ø/ should be relatively easier, as it has no L1 equivalent (English has no front rounded vowels) and thus will be categorized as a 'new' vowel. In contrast, the acoustic realization of French /e/ overlaps with that of English /i/. English learners may categorize such realizations as instances of their L1 /i/ and consequently fail to establish a new, target-like phonetic category for French /e/. Finally, some researchers have argued for the role of functional constraints including frequency, perceptual saliency, and articulatory ease (e.g., Colantoni & Steele, 2007a; 2008). Structures that are more frequent, more salient, and/or less variable may be less difficult to acquire.

Illustrative study 1.3. Colantoni & Steele (2007a)

Overview: These researchers studied the relative difficulty of aspects of the production of French <r>, namely voicing and manner. Difficulty was predicted using the variables of perceptual salience and relative articulatory difficulty of the two parameters, controlled for position in the syllable and word.

Languages: French (TL); English (L1)

Dependent variable: accuracy in realizing both manner and voicing of the French rhotic /ʁ/

Independent variables: *Learner*: (1) French proficiency [intermediate, advanced]; *Linguistic:* (2) Position of segment in the syllable and the word; [initial, final] in both cases

Hypotheses: *General*: French /ʁ/, a new segment for English-speaking learners, will be acquired in positions that favour the realization of the phonetic parameters of voicing and manner. *Specific*:(1) Manner versus voicing: due to the salience of /ʁ/'s manner and the articulatory difficulty of realizing voicing in a dorsal fricative, learners will overall be more accurate in realizing manner; (2) Voicing by syllable and word position: voicing of /ʁ/ will be acquired earlier in those positions that favour its phonetic implementation, namely, intervocalic position first, followed by word-initial position, then in word-final position, and finally in word-medial, syllable-final position; (3) Voicing in singleton versus voiced stop-rhotic clusters: voicing will be acquired in the former context first.

Methodology

Participants: 20 North American English-speaking learners (10 intermediate; 10 advanced); 10 native speaker controls (5 each of European and Quebec French).

Tasks: (1) French word-reading task: target stimuli were embedded in the French version of the carrier sentence 'I say [STIMULUS] again'. Each participant read the set of sentences three times in random order; (2) French passage reading: participants read the French version of *The North Wind and the Sun.*

Stimuli: For task (1) word reading: stimuli contained 14 singleton rhotics and 24 stop-rhotic clusters varying in position in the syllable and word. 34 distracters were included; (2) Passage reading: the text included 21 singleton rhotics.

Data analysis: (1) /ʁ/ manner and voicing: for each of the target stimuli from both tasks, an acoustic analysis was undertaken to measure the learners' realization of voicing (% laryngeal voicing of the entire segment) and manner (visual inspection of the waveform and spectrogram); (2) learner proficiency: to evaluate proficiency, the 20 learner readings were intermixed with those of five of the controls, randomized, and presented to three native speaker judges. The judges had to assign a score from '1 – Definitely non-native; strong foreign accent' to '5 – Definitely native; no foreign accent'.

Main findings

(1) Manner versus voicing: overall, learners of both proficiency levels were more accurate with manner than voicing, as predicted. Whereas the intermediate learners were already fairly accurate with manner, only a small number of the advanced learners had accurate voicing across positions.

(2) Voicing by syllable and word position: the hypothesized order of acquisition was only partially supported. The control data did not support such a hierarchy. Rather, there was a clear asymmetry between onsets and codas with the former being more voiced overall. The advanced learners' production was characterized by a similar asymmetry. In contrast, the intermediate learners' production was characterized by the lowest rate of voicing in onsets, contrary to predictions. This was due to these less-proficient learners realizing a greater percentage of fricatives in onsets, a manner which disfavours the realization of voicing.
(3) Voicing in singleton versus voiced stop-rhotic clusters: contrary to the hypothesis that learners would be more accurate with singleton /ʁ/, no differences were observed at the group level.

1.3.4 Variability

Whereas the study of developmental sequences and relative difficulty focuses on phenomena common to all L2 learners, one of the most striking characteristics of interlanguage perception and production is the large amount of variability that exists, both between learners (**interlearner** variability) and with the same learner (**intralearner** variability) at a given point in time or at two different points in time sufficiently close together that one does not believe such differences to be due to perception/production representative of two different stages of acquisition. Along with explaining CLI, developmental sequences, and ultimate attainment, among other phenomena, a complete theory of L2 acquisition must be able to account for such variability. Researchers have thus asked, *What are the sources of widely attested inter- and intralearner variability?* To date, two main types of factors have been shown to play a role, either linguistic and cognitive or, in the case of interlearner variation alone, learner-related.

Most work on L2 phonological variability has focused on the former. In the case of linguistic factors, it has been demonstrated time and time again that both inter- and intralearner differences in perception and production correlate with a variety of linguistic variables including segmental and prosodic context (e.g., Bayley, 1996; Hansen, 2004; Levy & Strange, 2008), word shape (e.g., Broselow, Chen & Wang, 1998; Steele, 2002), lexical category (Saunders, 1987), and the presence versus absence of orthography (e.g., Young-Scholten, 1995; Escudero & Wanrooij, 2010; Rafat, 2011). These factors, particularly segmental and prosodic context, are the same variables that linguists have shown to condition both allophonic and sociolinguistic variation. A smaller body of work has proposed that cognitive factors, including the extent to which deviance from the target will impede an interlocutor's comprehension (e.g., Weinberger, 1994; Abrahamsson, 2003), are also a source of variability.

In the case of interlearner variability, research on L2 acquisition has demonstrated that a relatively large set of variables correlate with interlearner differences including learner-external factors (e.g., quality and quantity of input, opportunities for language use) and learner-internal factors, that is characteristics of the learner him/herself. The latter, which are referred to as **individual differences**, include both personal characteristics such as motivation, extraversion, and self-efficacy; and cognitive abilities including phonological awareness and phonetic encoding abilities, analytical reasoning, and phonological memory (see Dörnyei & Shekhan, 2003 for an extensive overview). In previous empirical studies, some of these factors have been shown to correlate with L2 phonological variability, in particular age at onset of acquisition (e.g., Flege, Munro & MacKay, 1995a; Abrahamsson, 2012) and quality and quantity of input including the amount of explicit instruction (e.g., Moyer, 1999; Flege & Liu, 2001). As we will discuss in Chapter 3, given the importance of both linguistic-cognitive and learner variables in explaining interlanguage variability, it is of the utmost importance that researchers control for these variables when designing an experiment and analyzing the data obtained.

Illustrative study 1.4. Cardoso (2011a)

Overview: This study investigated the extent to which a variety of learner and linguistic variables affect L2 learners' perceptual accuracy of word-final stops.

Languages: English (TL); Brazilian Portuguese (L1)

Dependent variable: Percentage of accurately perceived word-final stops. In Brazilian Portuguese, underlying syllable-final consonants are realized phonetically with an epenthetic [i] and thus word-final stops are a new structure for such learners.

Independent variables: *Learner*: (1) Proficiency [no English knowledge, beginner, intermediate, advanced]; *Linguistic*: (2) Stop place of articulation [labial, coronal, dorsal]; (3) Quality of preceding vowel [tense, lax]; (4) Word size [monosyllabic, polysyllabic]

Research questions: (1) How does the perception of English codas evolve in the development of an L2?; (2) What factors contribute to the variable patterns observed in the process of acquisition?; (3) How systematic is the variation observed in L2 speech perception (e.g., is it gradual, and influenced by general linguistic principles such as word minimality restrictions?) (pp. 434–435)

Methodology

Participants: 51 Brazilian Portuguese-speaking learners of English (6 with no English knowledge, 6 beginner, 23 intermediate, 16 advanced). Proficiency was determined based on total classroom exposure to English (primary criterion), placement within participants' school proficiency system, and a background questionnaire. 3 native speaker controls participated in a pilot test of the task and stimuli.

Task: A forced-choice phone identification task in which participants had to choose one of three possible orthographic responses (Portuguese equivalents of 'consonant', 'vowel', or 'I don't know') when presented aurally with each of the stimuli.

Stimuli: Targets included 108 pseudowords (e.g., *fip* /fɪp/, *cipook* /sɪ.ˈpuk/) controlled for the linguistic variables mentioned above; preceding tense and lax vowels were /i,u,ɑ/ and /ɪ,ʊ,ɛ/ respectively. Furthermore, all target final stops occurred in a stressed CV<u>C</u> syllable. Vowel final distracters were included. All stimuli were recorded by a male native English speaker.

Data analysis: The percentage of accurate responses was calculated for the target stop-final stimuli.

Main findings

(1) The native speaker controls performed at ceiling (mean 97.5% identification accuracy).

(2) For the L1 Brazilian Portuguese speakers, regression analyses revealed that proficiency, stop place of articulation, and preceding vowel quality all contributed to the between-learner variation observed. Specifically, final stops were more likely to be perceived accurately (i.e., without an illusory final vowel) (1) with more proficient learners (intermediate and advanced); (2) when they were labial or coronal; and (3) when preceded by a lax vowel or /ɑ/.

1.3.5 Ultimate attainment including critical/sensitive periods for language acquisition

In the absence of general cognitive or linguistic-specific deficits, L1 acquisition results in children's mastery of the target language. In contrast, the vast majority of adult L2 learners fail to acquire native-like competence with many aspects of the TL. Indeed, some researchers have proposed that truly native-like attainment is impossible with learners for whom acquisition begins post-puberty (e.g., Lenneberg, 1967; Scovel, 1988; Long, 1990; Abrahamsson & Hyltenstam, 2009).

Non-target-like attainment is often said to be most evident in the domains of phonetics and phonology. While native-like performance has been observed in a limited number of cases (e.g., Escudero & Boersma, 2002, 2004; Escudero, 2009; Escudero, Benders & Lipski, 2009 for perception; Bongaerts, Mennen & Slik, 2000; Colantoni & Steele, 2006; Birdsong, 2007 for production), most L2 learners, even at their most proficient, have some degree of foreign accent. This widely observed characteristic has lead researchers to ask the question, *Is adult L2 speech* *learning subject to a critical/sensitive period?* While the most immediate and intuitive answer might be 'yes', if one understands 'critical period' in the strictest sense of Lenneberg's (1967) Critical Period Hypothesis – that is, that there is a period that ends towards 10–13 years of age, after which, due to neuro-biological maturation, L2 acquisition is fundamentally different in some respects and native-like attainment is impossible – there are at least three reasons to answer in the negative (see Flege, 1987a for other reasons). First, there are some adult L2 learners whose performance, as evaluated either by native speaker judges or instrumentally by phoneticians, is native-like (see above references). Secondly, and more importantly, previous studies, particularly those by James Flege and colleagues (e.g., Flege, 1987a; 1995; Flege, Munro & MacKay, 1995a) have demonstrated that the correlation between age at onset of acquisition and L2 phonological attainment is linear involving a gradual and continual decline over the lifespan. Were a critical period as conceived by Lenneberg to exist, one would not expect such a relationship; rather, there should be a bimodal distribution with those for whom learning began before puberty attaining native-like pronunciation and those starting later not doing so. Thirdly, L1 and L2 acquisition almost always differ with respect to one important variable, namely quantity and quality of input. Failure to attain native-like phonological competence may be due in great part to adult learners having access to relatively impoverished input vis-à-vis children (see Flege, 2009 for discussion). In summary, while all previous research supports the idea that – if all other variables are kept the same – 'younger is better', it is unclear that the traditional conception of a critical period is tenable, even for L2 phonological acquisition.

Accordingly, a potentially more fruitful avenue of research seeks to answer the question, *What factors best predict the degree of mastery of the target language?* The factors in question are, for the most part, those discussed in §1.3.4 with reference to interlearner variability. The most important of these is age at onset of acquisition, for which there is a negative relationship with ultimate attainment (see Ioup, 2008 for an overview). Other variables that have been demonstrated to correlate with ultimate attainment include length of residence in an L2-speaking country, formal instruction, degree of motivation to acquire the TL, general language learning aptitude, and amount of L1 use (see Piske, MacKay & Flege, 2001 for a review).

Illustrative study 1.5. Birdsong (2007)

Overview: This production study investigated the possibility of native-like attainment of both segmental and global phonetic proficiency, including the correlation between both.

Languages: French (TL); English (L1)

Dependent variables: (1) Production accuracy of vowel duration and voice onset time (VOT, measured acoustically); (2). Global accuracy (measured via native speaker judges' scores)

Independent variables: *Linguistic*: For the reading task, (1) vowel [i,e,o,u] or (2) consonant-vowel [p,t,k] in stimulus

Research questions: (1) Can late Anglophone learners of L2 French produce native-like consonants and vowels that are acoustically quite similar in the two languages? How rare (or how frequent) is nativelikeness at this level?; (2) Can late Anglophone learners of L2 French produce native-like French at the global level? Among subjects at the L2 end state who are not screened for French proficiency, what is the incidence of nativelikeness at the global level of analysis?; (3) Does nativelikeness at one level predict nativelikeness at the other level? (pp. 101–102)

Methodology

Participants: (1) Learner group: 22 native English speakers who had arrived in France at 18 years of age or later and who had resided in the Paris area for a minimum of 5 years; (2) Control group: 17 adult native French speakers

Tasks: (1) Word list reading (for segmental stimuli); (2) Short passage reading (for global phonetic proficiency)

Stimuli: (1) (a) Vowel duration: for each of the target vowels, two monosyllabic and one bisyllabic words in which the vowel occurred in an open syllable in stressed position; (b) For each of the voiceless stops, three words in which the stop was word initial but preceded by a vowel-final determiner; (2) Global proficiency: three short passages of two to three sentences from a literary work

Data analysis: (1) Vowel duration and VOT were measured acoustically; (2) Global proficiency was determined by presenting all of the L2 learner and native speaker texts to three native speakers, intermixed and randomized, who assigned a score from '1 – Definitely non-native; strong foreign accent' to '5 – Definitely native; no foreign accent'. Mean values were then calculated for all measurements.

Main findings

(1) (a) Vowel duration: as a group, the L2 learner values were longer than those of the native speaker controls. However, for three of the 22 learners, there were no differences vis-à-vis the mean control value using the criterion that the values needed to be within one standard deviation (s.d.) of the control mean; (b) VOT: the group learner means were longer than those of the controls. Using the same criterion for nativelikeness as with vowel duration (i.e., values within 1 s.d. of the control mean), nine of the 22 learners' VOT realizations were native-like.

(2) Global phonological proficiency: three L2 learners' readings received scores within the range of those assigned to the native speaker controls (4.5–5.0).

(3) Correlation between segmental and global proficiency: two of the L2 learners had native-like production as measured by vowel duration, VOT, and global proficiency. Moreover, most of those individuals who received high global ratings were native-like in their production of vowel duration and VOT. However, for at least two learners who received a high global rating, there were low levels of nativelikeness with the acoustically measured phonetic parameters.

1.3.6 The role of literacy and orthography in L2 speech learning

The overwhelming majority of participants in published studies of L2 phonological acquisition are (highly) educated and, thus, literate learners. Moreover, research has demonstrated that literacy affects an individual's native language perception (see work by Burnham and colleagues including Burnham, 2003; Horlyck, Reid & Burnham, 2012), phonological awareness (e.g., Him Cheung & Chen, 2004; Tyler & Burnham, 2006; see Koda, 2007 for an overview of the L1 literature on this topic), and, potentially, phonological representations (Port, 2007). Consequently, it is of interest to study the ways in which literacy and written language shape, both positively and negatively, L2 learning (see Tarone, Hansen & Bigelow, 2013 for a general overview), particularly given that, in contrast to child learners who spend the first years of their learning experience with access to auditory input alone, most adult learners encounter written and auditory input simultaneously from the onset and over the course of acquisition (Young-Scholten, 2002).

In the research undertaken to date, one can identify at least two questions related to L2 phonological acquisition and literacy: *How does literacy both facilitate and hinder non-native speech perception and production? What is the particular role of orthography?* A partial answer to the first of these questions must consider the fact that speech perception is multimodal with listeners integrating auditory and visual cues including orthography when creating percepts of the intended targets sounds (see Rosenblum, 2005 for an overview). It is also the case that phonological awareness, which is affected by orthography, is one of the prime components of general language learning aptitude (see, e.g., Sparks, Patton, Ganschow & Humbach, 2011). Consequently, in the case of literate learners who most often serve as subjects for published research, it is of interest to be able to determine what role, if any, literacy including its heightening of phonological awareness may have in shaping L2 learners' perception and production. The response to the second question is that orthography shapes both L2 learners' perception and word recognition in both positive and negative ways (e.g., Detey & Nespoulous, 2008; Escudero & Wanrooij, 2010; Escudero, Simon & Mulak, 2014; Showalter & Hayes-Harb, 2015; but see Simon, Chambless & Alves, 2010; Escudero, 2015 for potential limitations on orthography's effect). It also affects production (see Bassetti, 2009 for an overview) including by allowing learners to have contrastive underlying representations for vowels that they fail to distinguish in perception (Escudero, Hayes-Harb & Mitterer, 2008; but see Cutler (2015) for a demonstration of the profound negative effect of having distinct lexical items that an L2 learner cannot distinguish auditorily), by promoting negative effects of L1-based CLI (Rafat, 2011; 2015; Bassetti & Atkinson, 2015; Young-Scholten & Langer, 2015), by influencing the rate of epenthesis (Young-Scholten, 1997; Young-Scholten, Akita & Cross, 1999), and by shaping learners' establishment of underlying representations (Young-Scholten, 2000; Bassetti, 2006; Steele, 2009). We will discuss some of the specific orthographic effects in Chapters 4 through 8 when reviewing previous research on a range of themes and structures.

Illustrative study 1.6. Detey & Nespoulous (2008)

Overview: This study investigated the effects of orthography on the parsing of different types of consonant clusters.

Languages: French (TL); Japanese (L1)

Dependent variable: Mean rate of perceptual epenthesis. Following previous research by Dupoux, Kakehi, Hirose, Pallier & Mehler (1999), it was expected that some of the clusters would be perceived as involving a medial epenthetic vowel (e.g., /sC/ heard as /s<u>V</u>C/; see §7.4.1.2 for discussion of perceptual epenthesis among L2 learners).

Independent variables: *Linguistic*: (1) Type of cluster [/s/-stop, /s/-non-stop, obstruent-liquid]; (2) Position of cluster in the word [initial, medial, final]; *Task*: (3) Stimuli presentation modality [auditory (A), audiovisual (AV), visual (V)]

Hypotheses: (1) Lower rates of epenthesis will occur with visual presentation of the stimuli (AV and V modalities); (2) The rate of epenthesis will be conditioned by the type of cluster

Methodology

Participants: 62 Japanese-speaking L2 learners of beginner and intermediate proficiency participating within their existing university-level French classes.

Stimuli: Nonsense words consisting of (1) targets: trisyllabic forms involving each of the cluster types (e.g., word-initial /plokama/); and (2) distractors: quadrisyllabic and pentasyllabic forms created via insertion of an epenthetic /u/ or /o/ in the target clusters (e.g., /plokama/ → /pulokama/). Auditory stimuli were recorded by one of the native speaker authors for playback during the experiment; visual stimuli were presented in large font on a white sheet of paper via an overhead projector.

Tasks: All learners were presented with the stimuli first in auditory, then in audiovisual, and finally the visual modality, and were asked to indicate the number of syllables they heard by circling the corresponding number of ellipses on a sheet. For the audiovisual modality, participants were told that there might be discrepancies between the auditory and visual presentation of the stimuli; the response sheet included a column where this could be indicated.

Data analysis: All cases where the number of syllables indicated by the learners exceeded that of the stimuli were noted as instances of epenthesis, then the epenthesis rates were calculated for the group as a whole.

Main findings

(1) For all cluster types, a higher rate of epenthesis was observed with the visual (AV, A) modalities in all three word-positions (e.g., means: A: 59%, AV: 67%, V: 77%). This was particularly true of word-final clusters.
(2) A limited number of relatively small differences in rate of epenthesis were observed (e.g., for the A modality: stop-/r/ 22%, stop-/l/ 33%).

1.4 Chapter summary

In this first chapter, we have discussed the major research themes that underlie work on L2 speech, both for the perception and production of segmental and prosodic phenomena. These include the ways in which a learner's L1 and, potentially, other non-native languages interact with their interlanguage; how L2 perception and production change over time and what factors (dis)favour such change; the extent to which non-native hearer-speakers may master phonetic and phonological properties of the input; and how literate learners' knowledge of written language, particularly orthography, shapes their language learning and use. Keeping all of these major themes in mind as well as the specific research questions reviewed in this chapter, in Chapter 2 we turn to the theoretical frameworks – principles, models, and theories – that researchers have used in the experimental investigation of L2 speech.

1.5 Keywords

At the end of each chapter, we will highlight the most important concepts discussed. Once you have read each chapter, you should be able to define and explain each of these.

foreign accent, segments, prosody, phonetics, phonology, coarticulation, second language (L2) acquisition, third language (L3) acquisition, target language, input, interlanguage, cross-linguistic influence, developmental sequences, fossilization, end state, ultimate attainment, transfer or interference, variability, markedness, intra- and interlearner variability, individual differences

1.6 Review questions

These questions will allow you to review some of the main themes in the chapter. Answers can be found on the book website.

(1) Identify the acquisition phenomena illustrated by each of the following brief scenarios:
   (a) After five years of intensive immersion in Buenos Aires, Argentina, an L2 learner's pronunciation of the Spanish rhotic /r/ still differs from that of local native speakers and no further progress seems possible.
   (b) In a study on the acquisition of fluency in L2 learners of Arabic, a researcher discovers that it is possible to allocate learners to one of two groups in terms of their mimicry abilities and amount of previous target-language immersion experience.
   (c) An ESL (English as a second-language) teacher in Sydney, Australia, recognizes that, whereas her beginner Mandarin-speaking learners delete the final consonants of words such as *big*, *tough*, and *exact*, intermediate-level learners realize such words with a final schwa (e.g., *big*[ə], *tough*[ə], and *exact*[ə]), and the most advanced speakers' pronunciation is target-like.
   (d) In a café in Montreal, Jean-Marc notices that his new friend Jonathan, who just moved from Vancouver, sometimes calls the waiter [gaʁsɔ̃], like a native French speaker, and other times [gaɹsɑn], as though he were speaking English.

(2) What differences in input quality and quantity exist between L2 learners who acquire the TL in a formal learning context such as a language class and those whose acquisition takes place in a naturalistic setting such as while working? What consequences might these differences have for the speech perception and production of both types of learners?

(3) For your native language or a language that you have learned as an L2, choose a segmental or prosodic phenomenon, then
   (a) List typical examples of non-native perception or production. Where possible, identify errors that are likely due to cross-linguistic influence.
   (b) Identify a potential developmental sequence in the acquisition of the phenomenon by comparing the perception or production of beginner, intermediate, and advanced-proficiency learners.
   (c) Identify any ways in which learners of similar proficiency may differ in their perception/production. Try to suggest cognitive or affective variables that might correlate or explain such variability.

(4) In which ways may orthography help and hinder L2 speech learning?

1.7 Further reading

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2

Theoretical concepts and frameworks

Second language instructors as well as students approaching the study of second language acquisition are usually fascinated by the differences between native and L2 speakers. In pedagogical and applied linguistic texts, it is not uncommon to find detailed lists describing the ways in which the segmental and prosodic aspects of L2 speech differ from TL norms. Although such information is useful, it does not allow us to explain the behaviour of L2 learners. Having read Chapter 1, you are now familiar with many of the major questions that people have asked and continue to ask concerning L2 phonetic and phonological learning. Based on your own experience as an L2 learner or as someone who has regular contact with non-native speakers of your L1, you may have your own set of questions for which you would like to find answers. The goal of L2 speech research is to answer such questions. Setting out to do so may seem intimidating – where should you start? What is the best method for coming up with valid answers? Fortunately, we are almost never starting from zero. As is the case with all scientific research, it is rare to investigate a topic that has not been previously studied to some extent, perhaps with another language pairing or a different segmental or prosodic structure. The sum of all previous research is a treasure trove for researchers – as the physicist and mathematician Isaac Newton said, like many others before him, we stand on the shoulders of giants. In simple English, we have access to the reflections and research findings of others that can be used to structure and guide our own scientific investigation. As we will see in this chapter, one of the main ways in which this is done is through the theoretical concepts and frameworks that previous researchers have developed to formalize their own intuitions and research findings as well as those of others.

This chapter contains three major sections. We begin in §2.1 by asking three seemingly simple but important questions – *What is a theory? When and why should we use theory?* and, directly pertinent to our interests, *What does a theory of L2 speech learning need to account for?* In §§2.2 and 2.3, we discuss the major theoretical concepts and models that have been borrowed from L2 acquisition more generally or developed for L2 speech research on perception and production respectively.

2.1 Theory and L2 speech research

Most of the research to be discussed in Part III (Chapters 4 through 8) involves theoretical concepts and frameworks applied to specific structures. In the same way that it is important to know what theory to adopt when you want to answer a particular research question about vowels, consonants, or prosody, it is important to be able to evaluate competing theories' strengths and weaknesses. To do this, you need to have a set of principles in mind – you need to know what a theory should look like (§2.1.1). It is equally important to know when theory is unnecessary. As we will discuss in §2.1.2, sometimes it is important to start with purely fact-finding research because we simply do not know enough about the basics of the L2 speech phenomenon to describe it in sufficient, accurate detail. Finally, if we are interested not only in using theory to study L2 speech but also in applying our findings in turn to refine and elaborate theory, we need to know what a theory of L2 speech learning must include (§2.1.3).

2.1.1 What is a theory?

In the scientific sense, a theory is a **formal model** that seeks to account for one or more phenomena observed by researchers. It is based on a set of widely accepted principles and is falsifiable, that is, there are data that we can use to test it, to either support or, perhaps more importantly, refute it in part or in whole. When parts of a theory are falsified and thus untenable, researchers then revise the theory and test it further.

The quality of a theory may be evaluated in terms of its **adequacy**, which refers to the degree to which it provides a satisfactory account of the phenomenon it models. Researchers distinguish between three types of adequacy, namely **descriptive**, **explanatory**, and **predictive**. Whereas descriptive adequacy requires that a theory account for all of the known observations, a theory that is explanatorily adequate will also explain why things are as they are and not otherwise. Finally, the most adequate theories are also predictively adequate: beyond explaining what we have observed, they also allow us to predict what should and should not be possible beyond what we already know. To illustrate with an example relevant to L2 speech research, a complete, adequate theory of cross-linguistic influence in L2 phonological acquisition must (1) accurately describe which aspects of a learner's L1 and other previously acquired target languages do and do not influence both L2/L3 perception and production; (2) why the observed and unobserved influences (do not) occur; and, (3) for those phenomena for which the nature of CLI has not been previously observed including for a particular L1–TL pairing, what types of influence should and should not occur.

As you might imagine, building such a theory is highly complicated and requires the concerted effort of many researchers over many years. Fortunately, within the L2 speech research community, there is very much a sense of collaboration. As we will see in the remainder of this chapter, in some cases, researchers have developed and continue to develop theories particular to various aspects of L2 speech with different aims in terms of descriptive, predictive, and explanatory power, and with different levels of detail in their proposals.

2.1.2 When and why should we use theory?

With fairly broad strokes, we can group research into two main categories based on its general nature – normally, studies are primarily either **descriptive** or **analytical**. Of course, some research may be both. A descriptive study seeks to increase our understanding of the nature of a given phenomenon (e.g., the effect of orthography on production; the ways in which cross-linguistic influence changes over the course of acquisition; the effect of learners' L1 on the ways in which they pass through developmental stages). This is done in one or more ways: (1) by verifying the findings of previous research; (2) by exploring known phenomena in greater detail (for example, by increasing the range of linguistic variables manipulated or by increasing the range of TL proficiency among participants); or (3) by investigating phenomena for which little is known. While such a study will be motivated by a set of research questions (e.g., *To what extent are segmental production errors related to mismatches between L1 and TL orthography? Is CLI less common as learners become more proficient in the TL? Do learners with different L1s always follow a proposed developmental path?*), no hypotheses will be proposed and tested. Rather, the study will collect a set of data in order to answer the questions of interest. Stated somewhat simply, the primary goal of such studies is to expand our understanding by increasing the data available.

In contrast, while an analytical study will also be driven by a set of research questions and will seek to answer these questions based on a data set, it will go beyond asking a set of questions and will make a set of hypotheses following from one or more existing theories or principles. Theories are useful for structuring research in that they synthesize much of what is known about a particular phenomenon and, if they are predictively adequate, restrict the possible hypotheses that you can propose. As we have yet to discuss any particular theories, it is difficult to provide an easily understood example of a theory-driven study at this point. However, as we continue in this chapter, many concrete examples will be provided.

If you wish to undertake a descriptive study, most often because too little is known of a given phenomenon of interest, then no theoretical framework is necessary. This is not to say that such a study will be unstructured. Indeed, any scientifically valid experimental study, whether descriptive or analytical, will respect a set of methodological principles (e.g., choice of participants, task(s), and stimuli based on a series of well-established variables; quantitative and/or qualitative data analysis) to be discussed in detail in Chapter 3. Such work is extremely important, as it allows for expanding the empirical base and, often, provides new, more detailed insight into the true nature of L2 speech learning. Moreover, some research will not benefit from adopting a theoretical model – it may even be hindered if the theory is 'tacked on' in a way that only confuses the main issues. Unfortunately, there is sometimes pressure to 'be theoretical' on the erroneous assumption that analytical studies are superior to their descriptive counterparts. The irony is that any seasoned researcher can point to analytical studies whose interest diminishes with time as the theories themselves change or, in some cases, are abandoned. In contrast, good data stand the test of time and can be used again and again to test different theories.

2.1.3 What does a theory of L2 speech learning need to account for?

No complete theory of L2 speech learning exists that is capable of accounting for both perception and production. However, one can spell out what such a theory would need to account for. As a starting point, consider the partial model of segmental acquisition proposed in Colantoni & Steele (2008) and reproduced in Figure 2.1, which incorporates many of the principles of L2 speech learning discussed in Chapter 1, and various aspects of existing perceptual theories and principles from phonetic production research to be discussed in §§2.2 and 2.3 of this chapter.

In this model, the L2 acquisition of a TL segment is driven by the input to which a learner is exposed. This input is analyzed via a comparison of the learner's L1 and the TL, taking into account both acoustic and **phonotactic** information (i.e., distributional properties of segments within the syllable and word). Using the framework of the Speech Learning Model (see §2.2.1 for details), the model proposes that such a comparison results in the classification of a TL segment as either *old* (i.e., already existing in the learner's L1 segmental inventory), *similar* to one or more existing L1 segments, or *new*, that is, absent from the L1 inventory. Based on such comparison over time, the learner will establish a cognitive representation of the segment (also called a **category** in most models of L2 speech).

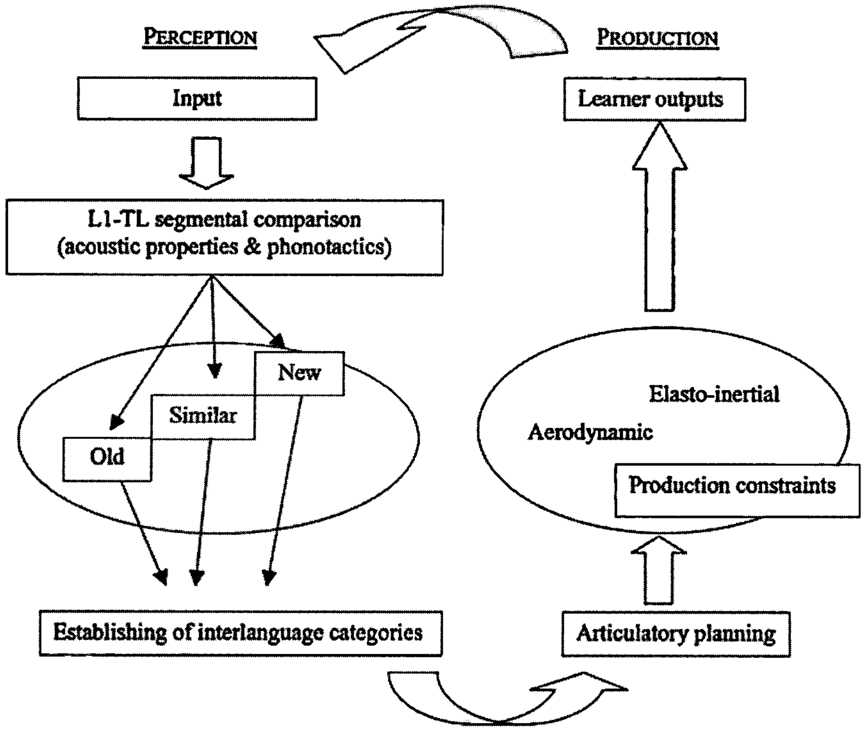

Figure 2.1. A model of L2 segmental speech perception and production (Colantoni & Steele, 2008).

Such a representation will feed articulatory planning. Articulation itself will be shaped not only by the existing L1-influenced articulatory patterns but also by universal production constraints on the airstream (aerodynamic) and articulators (elasto-inertial) resulting in the learner's real productions (outputs), which may be more or less native-like. These productions also constitute a type of input that feeds back into the system, given that speakers monitor their own speech.

While incomplete and restricted to segmental phenomena, such a partial model does indeed provide a framework for exploring some of the L2 speech research questions discussed in Chapter 1 including *How does input trigger speech learning?* and *How do previously acquired languages, both the L1 and other L2s, affect learners' perception and production?* However, it is unclear whether it is appropriate for the exploration of other questions such as *What are the consequences of differences in input quantity, quality, and modality for L2 speech learning*? and *What factors best predict the nature of CLI?* As with L2 acquisition in general, it is hard at present to imagine any single theory that could account for the wide range of phenomena that researchers study. Indeed, much work remains to be done.

One important aspect of L2 speech that a theory needs to account for is the underlying knowledge and mechanisms that lead to the procedures outlined in Figure 2.1. These involve theoretical assumptions concerning the general phonetic and phonological principles that underlie speech perception and production. A theory of L2 speech will draw on previous theoretical proposals from various disciplines including linguistics, psychology, and the speech sciences, among others, in order to answer questions including but not limited to *How exactly is the input, a continuous speech signal, processed into discrete segmental and prosodic categories by native speakers? Is this processing different in L2 learners? How exactly are the cognitive representations that underlie speech perception and production derived from the signal? How do such representations inform articulatory planning and the resulting acoustic properties in production?*

Keeping these various aspects of theory and theory use in mind, we now turn to models and theories of L2 perception (§2.2) and production (§2.3) separately. Over the years, theories have commonly concentrated on explaining one of these abilities more than the other, while others have only focused on one ability alone. Some of the reasons why L2 speech theories have focused on perception or production are described in the next section.

2.2 Theory and L2 speech perception

It is widely recognized that L2 speakers normally have a foreign accent, that is, they sound different from native speakers of the TL. As mentioned in Chapter 1, the most salient evidence of foreign accent is found in L2 learners' production. For example, as we discussed in §1.3.2, European French-speaking learners of English produce English /θ/ and /ð/ in words such as *think* or *this* as their L1 /s/ and /z/ respectively (i.e., as [s]ink, [z]is). The ubiquitous nature of foreign-accented speech and the ease of documenting L2 learners' productions led some theories (Lado, 1957; Eckman, 1977; 1991; Major, 1987a) to concentrate on describing CLI in production and on explaining L2 performance in terms of production difficulties. As we will see in §2.3.2.2, these early theories proposed that learners find L2 sounds difficult to produce when they are absent from the L1 or typologically rare.

Although most early observations and theories of L2 speech were based on production data, perception-based theories of L2 speech emerged with the increasing availability of cross-linguistic speech perception research. In a review article, Strange (1995) mentioned that this cross-linguistic evidence clearly shows that L2 learners also have **perceptual foreign accents** (pp. 22, 39) – CLI also affects perception and results in learners having difficulty distinguishing TL categories. Within perception-based theories of L2 speech, this evidence also led to a common underlying assumption, namely that learners' difficulty with producing L2 sounds is likely to be perceptual (but see §2.3 for examples of aspects of foreign-accented speech that are arguably linked to constraints on production alone). This assumption is primarily based on the fact that a learner first perceives TL structures before attempting to produce them, as is the case in L1 acquisition where young children have extensive knowledge of their L1 perceptual system long before they start producing their first words.[1] However, adult learners start producing the TL earlier on in the learning process than do young children. Whereas children start producing words around their first year of life, adult learners commonly attempt to produce their TL quickly after first contact. A perception-based model may thus suggest that L2 learners' difficulty with the TL and resulting foreign accent may depend on whether learners are able to detect differences between their own productions and those of native speakers of the TL.

In L2 speech, although the interrelation between perception and production is not always straightforward, as we will see in Chapters 4 through 8, some studies have shown that learners' perceptual development precedes that in production, suggesting that learners may resolve their difficulties with perceiving L2 sounds faster than their production difficulties. Some perception-based theories even propose that learners need to develop their L2 perception before development in production can begin. Flege and colleagues (Flege, 1995; Flege, Bohn & Jang, 1997) claim that the ability to produce TL categories with native-like accuracy depends on the absence of perceptually based difficulty. Similarly, Escudero (2007) reviews a large body of research demonstrating the precedence of accurate perception over accurate production in L2 development, and argues for exhausting the perceptual basis for L2 speech difficulty before considering articulation-based explanations.

All perception-based theories rely on the notion of CLI to explain variable performance with different types of segments and contrasts, in that they propose that learners' difficulty with perceiving the TL depends on the types of correspondences between TL and L1 categories. Languages vary in their contrastive phonemes (e.g., English has /b/ and /v/ while Spanish only has /b/), but phonemes may vary across languages in terms of their phonetic properties (e.g., English /p/ is produced differently from French or Spanish /p/, as we will see in Chapter 5). Theories vary with respect to the **level of analysis** that they use to describe TL and L1 categories. Some use contrastive phonological information such as phonemes, features, or higher order gestural invariants, while others use non-contrastive sub-phonemic or

[1] In the case of phonology, there is, however, an important way in which child and adult learners differ: whereas adults begin the acquisitional process with a fully developed articulatory system, when children first encounter language there are aspects of their anatomy (e.g., the proportional size of their tongue to the oral cavity) that impede target-like speech production.

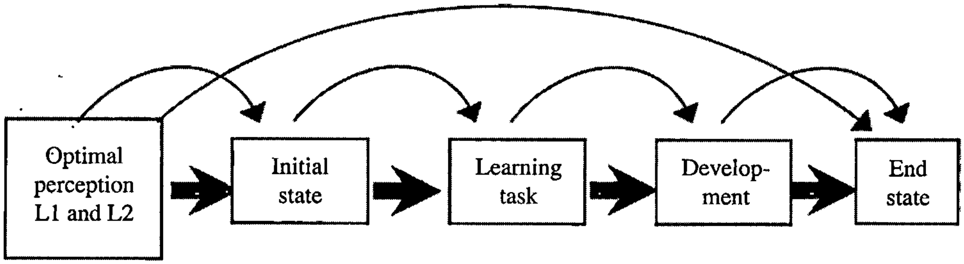

Figure 2.2. Proposed steps in L2 speech perception development (adapted from Escudero, 2005).

phonetic information such as acoustic cues or gestural phonetic information, but most of the theories that we will discuss below use a combination of the two. In this respect, the aim of L2 perception theories is to predict and explain L2 difficulty based on their particular formalization of CLI. In other words, each theory adopts a specific view regarding the phonological and/or phonetic properties of learners' L1 categories that influence the perception of TL categories.

Models of L2 perception provide testable hypotheses concerning degree of difficulty. These hypotheses are then used to inform research questions within analytical studies that are specific to the learners and languages involved (e.g., *Do Spanish-speaking learners perceive English or Dutch vowels as accurately as native listeners? Do such learners have difficulty with all TL categories or only with some?*). They also aim at predicting how interlanguage categories change over time and include predictions as to what learning scenarios are most likely to lead to successful acquisition. An illustration of a time path for L2 perception is shown in Figure 2.2. The figure shows that, depending on the differences between L1 and TL categories, two possible learning paths may take place. The first, illustrated by the long arrow extending directly from 'Optimal perception L1 and L2' to 'End state', depicts the case in which the L1 and TL categories are identical, resulting in native-like competence at both the initial and end states. Here, nothing new has to be learned, so target-like perceptual performance is immediate. In contrast, when the L1 and TL categories are different, the learning process will likely take more time and will involve a number of steps, as depicted by the many small arrows. It should be noted that the sequence of 'Learning task' followed by 'Development' does not represent a single occurrence but rather a process, and that 'Learning task' does not refer to the type of tasks used in experimental research but rather to the on-going analysis of the TL input followed (hopefully) by changes in learners' perceptual (and production) grammars.

In what follows, we will present the perception-based models that have inspired the majority of L2 perception studies and many production studies that have been conducted on a variety of L1–TL combinations, namely the Perceptual Assimilation Model (PAM; Best, 1994; 1995), the Speech Learning Model (SLM; Flege, 1995; 2003), and the Second Language Linguistic Perception model (L2LP; Escudero & Boersma, 2004; Escudero, 2005; 2009). To different degrees, these three models are both phonetic and phonological in nature, as they take into account phonetic detail but rely too on phonological theory for their explanations. We will also discuss a perception-based model informed purely by phonological theory, namely Brown's (1998) Phonological Interference Model (PIM). All of these models will be presented vis-à-vis some of the research questions that we discussed in Chapter 1 concerning the effect of CLI on L2 performance (§2.2.1), how L2 development takes place (§2.2.2), and L2 variability (§2.2.3).

2.2.1 Interpreting cross-linguistic influence and predicting levels of difficulty

As mentioned above, one common feature among L2 perception models is that they propose ways of predicting degrees of success related to the extent to which a learner's L1 and TL categories are similar/different. This approach has a long-standing tradition – it is at the core of Lado's (1957) Contrastive Analysis Hypothesis, which proposes that L1 'habits' are used in the process of learning an L2, with positive versus negative results for similar and different habits respectively. Lado's proposal targets the level of the phoneme: specifically, TL phonemes that are not present in the L1 should be more difficult to acquire. There is much evidence that this proposal is untenable. For example, Flege (1987b) showed that the opposite was true for American English-speaking learners of French, as they exhibited less difficulty with /y/, a French vowel that does not exist in English, than with /u/, a vowel shared by English and French. Similarly, following Lado's proposal that CLI applies at the phonemic level, one would expect that Japanese and Chinese-speaking listeners, neither of whose L1 has the English phoneme /ɹ/, would experience similar degrees of difficulty when distinguishing English /ɹ/-/l/. However, as Brown (1998) shows, only Japanese-speaking learners of English experience difficulty in distinguishing this contrast.

Brown's **Phonological Interference Model** (PIM; Brown & Matthews, 1997; Brown, 1998) accounts for the differential degree of difficulty that Japanese- and Chinese-speaking learners of English have with the /ɹ/-/l/ contrast by positing an alternative interpretation of CLI. Specifically, Brown proposes that the difference between Japanese- and Chinese-speaking learners is accounted for if one assumes that the phoneme is not the smallest unit in phonology, but that phonemes consist of **distinctive features** (e.g., [front] that contrasts /i/ from /u/ or [voice] that contrasts /p/ from /b/; see §4.1.1 for further discussion of features with respect to vowels). In other words, the influence of L1 categories does not occur at the level of phonemes but rather at the level of their constituent features. Brown argues that L2 learners perceive TL categories using L1 features that are **redeployed** or reused in L2 perception. According to this model, the absence of a distinctive feature, rather than of a particular phoneme, results in non-target-like L2 perception. For example, English /ɹ/ and /l/ are distinguished by the feature [retroflex], with the former being retroflex and the latter non-retroflex or [coronal]. A similar featural distinction is found among Chinese consonants: this language distinguishes between [alveolar] versus [retroflex] sibilants (e.g., /s/-/ʂ/), while Japanese does not have such a featural distinction. Within the PIM, the presence of [retroflex] in Chinese, absent in Japanese, explains why the latter but not the former learners of English have difficulty with this contrast.

Speech perception research challenges this model with respect to its possible predictions for L2 vowels. Brown's model would predict, for example, that native English speakers should find the French /i/-/y/ contrast difficult to discriminate: while French /y/ shares the feature [front] with English /i/, the feature [round] is not a distinctive feature of English. Consequently, according to the PIM, English speakers should not be able to distinguish /y/ from /i/, as they are both front vowels, while they should be able to accurately discriminate /y/-/u/ in the same way as they discriminate English /i/-/u/, and they should have less difficulty with learning French /u/, as it shares the same feature [back] as English /u/. However, Gottfried (1984) demonstrated that the opposite is true: English speakers confuse French /y/-/u/ but not French /y/-/i/ (see Chapter 4 for other studies with similar findings). Similarly, Brown's model would predict that French /u/ and /t/ would be easy to acquire for English learners, as the distinctive features involved in both categories, namely [back] and [coronal], are found in English, while French /y/ would be most difficult as English lacks [round] as a distinctive feature. However, Flege (1987b) found the opposite, namely that English-speaking learners performed better with French /y/ than French /u/ and /t/. These earlier findings seem to challenge Brown's feature-based model.

Flege's **Speech Learning Model** (SLM; 1995; 2003) formalizes perceptual CLI as a type of **equivalence classification**, which is a general phenomenon by which perceptual objects are equated with existing categories formed based on previous experience (Flege, 1987b, pp. 49–50). Experience with an L1 leads to a decreased ability to detect differences between TL and L1 categories: as speakers become highly skilled at classifying the various phonetic realizations of a given L1 category, it becomes more likely that they will use this L1-appropriate perception to parse all speech including instances of TL categories (p. 50). In cases where the L1 and TL differ, particularly in less-salient ways, L1-based perceptual mechanisms will lead to non-target-like performance. With the SLM, although the capacity to create L2 categories, which is essential for attaining native-like proficiency, remains available to adult L2 learners, equivalence classification may prevent or block L2 **category formation** (Flege, 2003). One of the model's main hypotheses is that the more similar a TL category is to an L1 category, the more likely it will be equated to an L1 category and the less likely it will be for the learner to form a new, target-like L2 category (Flege, 1995; 2003). That is, **similar TL categories**, those which are analyzed as existing L1 categories despite phonetic differences, should be more difficult to acquire than **new categories** that do not closely resemble any L1 category. This explains why English-speaking learners of French are more successful at producing the new French vowel /y/ than they are at producing French /u/ and /t/, which are similar to their English counterparts. As summarized in Illustrative study 2.1, this also explains Flege, Tagaki & Mann's (1996) finding that Japanese-speaking learners' perception of English /ɹ/ is better than that of English /l/. Importantly, in the SLM, the position of the TL categories within a word is highly relevant. Empirical evidence suggests that learners have different degrees of difficulty depending on whether the TL category occurs in word-initial, -medial or -final position (e.g., Strange, 1992 cited in Flege, 1995). This led Flege to propose that it is at the level of **position-sensitive allophones** that L1 and L2 categories are compared. As explained in Flege (1995), this level of analysis is not as abstract as the phoneme but nonetheless corresponds to an abstract representation in long-term memory. That is, TL categories are either similar or new in comparison to the closest position-sensitive allophone in the L1.

Illustrative study 2.1. Flege, Takagi & Mann (1996)

Overview: This study examined the effect of lexical familiarity and experience with the TL on learners' identification of words containing /ɹ/ and /l/.

Languages: English (TL); Japanese (L1)

Dependent variables: (1) Selection of lexical definitions (from five possible options); (2) Subjective frequency rating (scale from '1 = never heard or said' to '7 = very often heard or said'); (3) Consonant identification accuracy.

Independent variable: *Learner*: Experience with the English language [inexperienced, experienced learners].

Research questions: (1) Can experienced learners identify /ɹ/ and /l/ at a rate comparable to that of native speakers?; (2) Does subjective lexical familiarity have an effect on their level of accuracy?

Hypotheses: (1) Experience with the English language will allow Japanese-speaking learners to perform in a more native-like manner; (2) Learners will be influenced by the *relative subjective familiarity* of the words in a minimal pair involving /ɹ/ and /l/ (e.g., *red* is more familiar to the learners than either *rook* or *lead* (verb)).

Methodology

Participants: 36 participants, 12 in each of three groups (native speakers of English (NE); Japanese-speaking learners living in the US for 12 or more years (EJ); learners with three or fewer years living in the US (IJ)).

Tasks: (1) *Lexical task*: This assessed participants' familiarity with 42 words containing /ɹ/ and /l/ in word-initial position (e.g., *road, load*) and with four non-words (*rine, ruck, leck, lun*); (2) *Experiments 1 & 2*: Participants were asked to identify the initial consonants of English words when produced in the initial position of words or when extracted from the word and presented in isolation. Participants were asked to circle the letter that corresponded to the TL consonants on an answer sheet.

Main findings

(1) *Lexical task*: Both groups of learners made very few mistakes when choosing the correct definition of the words, but the EJ group had more correct definitions than the IJ group. The NE listeners demonstrated greater word knowledge and more certainty that the non-words were not part of the English lexicon than both groups of learners. EJ and NE subjects had higher levels of familiarity with the 42 words presented than the IJ group.

(2) The results of *Experiments 1 & 2* showed that the EJ group correctly identified stimuli containing English /ɹ/ and /l/ more often that did the IJ group, but, in most cases, less accurately than NE group. NE participants performed at ceiling and did not show an effect of lexical familiarity, while the Japanese learners' performance was influenced by the context in which the consonants were presented. Specifically, the EJ group had accuracy rates comparable to those of the NE group when identifying words from a minimal pair that they rated as having roughly the same familiarity (e.g., *rate – late*). Moreover, when lexical information did not play a role, as in Experiment 2 where /ɹ/ and /l/ were presented in isolation, extracted from their word or non-word contexts, the EJ group also had comparable accuracy to the NE group.

The results for the EJ group in Flege, Takagi & Mann (1996) show that learners can achieve native-like performance. According to the SLM, this accurate performance is due to the existence of new L2 categories for English /ɹ/ and /l/, indicating that category formation is available to learners across the lifespan. The model also suggests that, in cases where category formation is prevented by equivalence classification, learners can eventually detect differences that may lead to relatively good performance despite the absence of a new TL category. However, the process of detecting differences between TL and L1 categories may be slower and constrained by influences from both the TL and the L1. This is because similar TL and L1 categories are stored as a single category, called a **diaphone** or **merged category**, which results in **bidirectional CLI** between the TL and L1 categories. MacKay, Flege, Piske & Schirru (2001) demonstrate that Italian-speaking learners of English may have a single, merged voice onset time (VOT) category that they use to produce the voiced stops /b, d, g/ in both Italian and English. As will be more thoroughly explained in Chapter 5, these voiced consonants are produced with prevoicing in Italian, while they are characterized by short-lag VOT in English, which makes them sound like Italian [p,t,k]. These researchers show that learners with the most native-like English productions also had the most English-like values in Italian. In §2.2.2, we will further discuss the mechanisms that the SLM proposes to constrain L2 learning, one that applies to TL categories that are equated with L1 categories (similar categories) and another that applies to new TL categories. In addition to the effect of similar versus new categories, the SLM proposes that a number of learner variables underlie L2 speech difficulty, such as the age at which L2 learning began as well as the amount of exposure and daily use of each language, all of which will be discussed within the factors that contribute to variability in L2 learners' perception (§2.2.3). Although the predictions of the SLM for L2 difficulty are based on a comparison of individual TL and L1 categories, Flege (2003, p. 347) suggests that learning two TL phonemes that resemble two different allophones of the same L1 phoneme (e.g., both English liquids /ɹ/-/l/ share phonetic properties of Japanese /ɾ/) is more difficult than acquiring two TL phonemes (e.g., /t/-/d/) that resemble allophones corresponding to two different L1 phonemes (e.g., English [t] as in *stay* and English [d] as in *day*).

Best's (1994; 1995) **Perceptual Assimilation Model** (PAM) makes predictions concerning the difficulty of TL contrasts rather than individual TL phonemes as is done in the SLM. Although this model was developed to address not L2 learning but rather the perception of foreign or non-native contrasts by naïve listeners who have not yet been exposed to the TL (i.e., for cases of cross-linguistic perception), many studies have used this model to explain L2 speech perception. In addition, the model's predictions more recently have been applied to L2 learning (PAM-L2; Best & Tyler, 2007). As the L2 version of the model uses the same principles as the original model, we will discuss the PAM here and will explain the predictions for L2 developmental sequences suggested within the PAM-L2 in §2.2.2.

The main proposal of the PAM is in line with that of the SLM and PIM models described above, in that it argues that not all TL categories are equally difficult for non-native listeners. Specifically, the model proposes that difficulty in the discrimination of a TL contrast can be predicted based on how a TL phonemic contrast is *assimilated* to L1 categories. Unlike the SLM and PIM, the PAM explicitly rejects the assumption that categories are stored as mental representations. Instead, it follows a **direct realist** or ecological approach to speech perception that argues that listeners detect information regarding the **articulatory gestures** that generated the speech signal (Fowler, 1986; Best, 1995). Articulatory gestures are defined in terms of the organs (articulators such as the lips or tongue), constriction locations (place of articulation; e.g., labial, coronal, dorsal), and constriction degree (manner of articulation; e.g., stop, fricative) involved in speech production. Within the model, a phoneme is understood as a class of articulatory variants that have a similar function in a language: they distinguish lexical items and participate in context-conditioned allophony. CLI involves non-native listeners perceptually assimilating the articulatory gestures of TL phonemes to the articulatory gestures underlying those of their L1 phonemes.

The PAM thus predicts that a learner's ability to discriminate a TL phonemic contrast depends on the way in which each of the members of the contrast is assimilated. If there is **two-category assimilation (TC)**, where each TL phoneme involves articulatory gestures that correspond to those of a distinct L1 phoneme, discrimination is predicted to be excellent. In contrast, when there is **single-category assimilation (SC)**, where the two TL phonemes involve the gestures of a single L1 phoneme, discrimination will be poor. Alternatively, the two members of a contrast might assimilate to a single L1 phoneme, but one may be a better exemplar of the L1 phoneme than the other, leading to **category goodness assimilation (CG)**, where discrimination is predicted to be intermediate between SC and TC. Best & Strange (1992) showed that these differential predictions are borne out with Japanese listeners when presented with the English /ɹ/-/l/ and /w/-/ɹ/ contrasts, as the learners' perceptual assimilation results predicted their discrimination ability: they exhibited SC for the former (i.e., they classified English /ɹ/ and /l/ as equally poor exemplars of Japanese /ɾ/ or /w/), and CG for the latter (i.e., they classified English /w/-/ɹ/ as a good and bad exemplar of Japanese /w/ respectively), and were more accurate at discriminating the CG than the SC contrast.

The model includes three more possible assimilation scenarios: (1) the two members of a TL contrast may be assimilated to roughly the same extent to two or more TL phonemes, resulting in an **uncategorized–uncategorized** (UU) contrast; (2) one of the members may be assimilated to a single L1 phoneme and the other to two or more (**categorized–uncategorized** or UC); and, finally, (3) the members of a TL contrast may have articulatory features that do not resemble any speech sound and therefore are **non-assimilable** (NA). Despite the existence of these three other possible scenarios, most studies conducted within the PAM have concentrated on TL contrasts of the SC, TC, or CG types, as in Best, McRoberts & Goodell (2001; Illustrative study 2.2), which provides support for the prediction that TC contrasts will be easier to discriminate despite the fact that their members are not present in the L1, as in the case of the Zulu and Tigrinya (an Ethiopian language) contrasts for English-speaking listeners.

Illustrative study 2.2. Best, McRoberts & Goodell (2001)

Overview: This study tested the PAM's predictions of differential discrimination accuracy depending on how TL phonemes assimilate to L1 phonemes.

Languages: Zulu, Tirinya (TLs); English (L1)

Dependent variable: Discrimination accuracy (%).

Independent variable: Perceptual assimilation type: percentage of time a TL segment is identified as an L1 phoneme presented as a response option.

Research question: Are the PAM's predictions regarding differential levels of discrimination accuracy supported by the data?

Hypotheses: (1) Discrimination of a TL contrast should be close to 100% if the contrast is perceived as phonologically equivalent to an L1 contrast (two-category (TC) assimilation); while (2) discrimination should be lower but still very good if the contrast is perceived as a phonetic distinction between good and poor exemplars of the same L1 category (category goodness; CG); and, finally, (3) discrimination should be lowest if the two segments of a TL contrast are phonetically equivalent to the same extent to a single L1 category (single-category assimilation; SC). In summary, the prediction for levels of discrimination accuracy is TC>CG>SC.

Methodology

Participants: (1) *Experiment 1*: 22 native speakers of American English (AE) recruited from an Introductory Psychology subject pool, 15 females; (2) *Experiment 2*: 19 native speakers of AE, 10 females.

Stimuli: (1) *Experiment 1*: An adult female native Zulu speaker produced multiple tokens of each of six target consonants in CV nonsense syllables. The stimuli formed minimal pairs of consonants that were predicted to conform to three different assimilation patterns, namely lateral fricatives /ɮɛ/-/ɬɛ/ (TC), velar stops /ka/-/k'a/ (CG), and bilabial stops /bu/-/ɓu/ (SC); (2) *Experiment 2*: One other non-native contrast that was expected to conform to TC assimilation was tested alongside two native AE contrasts. A male native speaker of Tigrinya produced multiple tokens of /p'ɛ/-/t'ɛ/, while a male native speaker of AE produced multiple tokens of /sɛ/-/zɛ/ and /ʃɛ/-/ʒɛ/. The AE contrasts were chosen because they were among the spellings used for the Zulu lateral fricatives in Experiment 1 and were produced with the same active articulator as that contrast.

Tasks: In both experiments, participants performed two tasks: (1) An AXB discrimination task for each contrast: A and B were tokens of one of the two phonemes in the contrast and X was a different token of either phoneme. For each trial, listeners were asked to circle whether the X token was the same as the A or B token on an answer sheet. Each AXB test contained 96 trials in 12-trial blocks; (2) A task which involved writing down each of the syllables that they heard using English orthography, commonly known as a perceptual assimilation task. They were also asked to write any further qualitative description of the sound that they heard.

Data analysis: (1) Percentage correct performance was assessed (i.e., whether an X token was correctly matched with a token of the same consonant (A or B)); (2) Assimilation was considered SC if the two consonants in a contrast received the same spelling with no extra qualitative difference noted, while it was analyzed as CG if the two consonants in a contrast received the same spelling but received additional letters or descriptions to note a phonetic or acoustic discrepancy between the two consonants. When the two consonants received different spellings, the assimilation pattern was labelled TC.

Main findings

(1) *Experiment 1*: Results for Task 1 showed that the lateral fricatives (predicted TC) had higher discrimination accuracy than the velar stops (predicted CG) which had in turn higher accuracy than the bilabial stops (predicted SC). For Task 2, only 15 participants had English spellings that matched the predicted assimilation patterns for all three contrasts. The authors compared the discrimination accuracy for only these 15 participants and found that the results for accuracy were still TC>CG>SC, confirming the PAM-based predictions.

(2) *Experiment 2*: The authors found that discrimination for the two AE contrasts in Task 1 was at ceiling, while discrimination of the non-native Tigrinya contrast was more variable but still very high. In Task 2, 16 out of 19 participants provided different spellings for the two non-native consonants, which confirms that this contrast is indeed a TC-assimilation type and that, therefore, it yields high discrimination accuracy. The authors conclude that not all non-native contrasts are difficult and that difficulty indeed depends on the type of assimilation pattern.

As mentioned in Tyler, Best, Faber & Levitt (2014), most previous analytical studies that have been conducted by Best and colleagues to test the PAM's predictions have focused on non-native consonantal contrasts. Very few non-native vowel studies have interpreted their findings in terms of the PAM using the model's two required tasks of perceptual assimilation and discrimination (but see Polka, 1995; Polka & Bohn, 1996), and a sole recent study has tested all assimilation types for non-native vowels (Tyler, Best, Faber & Levitt, 2014). The PAM-L2 essentially reinterprets the perceptual assimilation patterns discussed above in the context of L2 learning and makes predictions for L2 patterns when a contrast involves SC, CG, or TC assimilation. As these predictions are for developmental sequences, they will be described in §2.2.2. In this chapter, we only briefly mention the tasks that are used to test the PAM model. Chapter 3 will include descriptions of a variety of perception tasks commonly used in L2 speech studies (§3.3.2.1), including perceptual assimilation and discrimination tasks, so that the reader is equipped with the relevant methodological knowledge to assess the findings of the studies that will be presented in Chapters 4 through 8.

The many studies conducted within the SLM and PAM reveal that there is a considerable degree of individual variability in the perception and production of L2 vowels and consonants. As we will see in §2.2.3, Flege and colleagues have considered a large array of extralinguistic factors including Age at Onset of Acquisition (AOA) and the amount of language exposure or use, that can explain differences among L2 learners. However, the SLM approach is to group learners under a specific extralinguistic variable (e.g, AOA: learners who started learning the TL early versus those who started late) prior to comparing their performance. Within work adopting the PAM as a theoretical framework, variability in perceptual assimilation patterns has been reported but not explained. In this respect, Tyler, Best, Faber & Levitt (2014) propose that individual assimilation patterns rather than group results should be at the core of a model that aims to explain L2 learning, as individual patterns may more accurately predict the development of each individual L2 learner. This is especially the case for vowels, as they tend to exhibit more variability in L1 and TL production and perception.

The **Second Language Perception Model** (L2LP; Escudero & Boersma, 2004; Escudero, 2005; 2009) was specifically designed to account for individual variation in non-native speakers of varying proficiency, including those who have not yet started to learn the TL as well as those that are highly advanced. The aim of the model is to predict L2 difficulty in each individual through a comprehensive comparison of their L1 categories and those of the specific variety of the TL that they will be (or are currently) learning. The L2LP model thus predicts a developmental sequence for each naïve listener and then tests whether the predicted development is indeed observed in beginner, intermediate, and advanced L2 learners with a shared L1. Although analyzing data from learners with different proficiencies is an appropriate way to test the L2LP's predictions, **longitudinal data** from a naïve listener who gradually gains L2 proficiency is more desirable. However, longitudinal L2 speech studies are rare due to the difficulty involved in testing L2 learners multiple times over longer periods of time (but see some examples of longitudinal L2 vowel experiments in Chapter 4). In general, the L2LP predictions are similar to those of the PAM, as the former model also considers TL phonemic contrasts rather than individual segments. Adopting SLM terminology, within the L2LP, TL contrasts can be **new contrasts** or **similar contrasts**, depending on whether the members of the TL contrast are heard as the same or two different L1 categories. Given the variable production of vowels in the input to L2 learners and the large variability in the performance of L2 learners reported in many previous studies, the model has concentrated on L2 vowels.

Rather than modelling CLI in terms of articulatory gestures as is done in the PAM, the L2LP follows the SLM in using acoustic information to predict cross-linguistic categorization patterns, which in turn are used to formulate developmental sequences. Vowels may be particularly difficult to describe in terms of similarities and differences in articulatory gestures, given the continuous nature of their production (Strange, 2007) that may be better captured by using continuous acoustic values. For example, due to the influence of the preceding consonant, or **coarticulation**, some tokens of French /y/ may be closer to English /i/, while others may be closer to /u/ in terms of how their second formant (F2) values, which indicate fronting, compare to those of the corresponding English vowels. The L2LP proposes that a native listener is equipped with a **perception grammar**, which is a system that weighs and parses incoming acoustic values, and ultimately maps these acoustic values onto perceptual representations such as allophones or phonemes. Native listeners are **optimal perceivers**, that is, they have a perception grammar that can efficiently parse continuous acoustic values onto abstract phonological representations. Given that the acoustic values of L1 categories and the listeners' perception grammars are directly linked, very detailed acoustic measures are needed for accurate prediction and description of a listener's perception grammar. Within the L2LP, acoustic analyses involve using the same methodology for data collection for both the L1 and TL, including using measures taken from a given token at different temporal points (e.g., vowel onset, midpoint, and offset), and running either computational modelling or statistical analyses for the comparisons. It is thus assumed that the acoustic values of a given individual's L1 categories are good predictors of how this listener perceives L1 categories and will perceive TL categories. For instance, Escudero, Sisinni & Grimaldi (2014) show that although Salento Italian (SI) and Peruvian Spanish (PS) have the same number of vowel phonemes, the vowels are produced in a phonetically different way in these two languages. Following the L2LP, these authors predict that SI and PS listeners will perceive Southern British English (SBE) vowels differently. A comparison of English vowel categorization data from the two groups of listeners shows that this prediction is borne out.

The L2LP predicts that the degree of variability in the acoustic values of L1 categories will inevitably impact a listener's native perception grammar, and consequently how s/he hears TL categories. For example, Escudero & Boersma (2004) showed that the English vowels /i/ and /ɪ/ have different acoustic values when produced in SBE versus Scottish English (SE). These researchers also showed that

L1 speakers of these two English varieties perceive synthetic stimuli of the two vowels differently and in keeping with the acoustic values of their own L1 variety. With respect to L2 perception, Escudero & Boersma showed that native Spanish speakers' performance followed different developmental sequences depending on which of the two varieties was the target. Spanish-speaking learners of SE perceived the TL vowels in a close-to-native-like manner, as the vowels of this variety have formant frequencies that are comparable to those of Spanish /i/ and /e/ respectively. In contrast, the Spanish-speaking learners with SBE as a TL distinguished the vowels on the basis of their duration differences. Given that the SBE but not the SE vowels are distinguished by different vowel duration and formant frequency values (SBE /i/ is both longer and higher than /ɪ/), the differential perception by speakers of the same native language found in this study shows that TL acoustic values indeed influence L2 learners' performance. Escudero & Boersma also show that the use of duration by Spanish speakers results from the absence of vowel duration categories in Spanish, as will be further explained in Chapter 4.

Other studies conducted within the L2LP have also demonstrated that individual differences in the perception and production of L1 categories determine and explain developmental sequences observed in L2 speech learning. For instance, Mayr & Escudero (2010) found that English-speaking learners of German performed differently depending on their individual perceptual assimilation patterns, as summarized in Illustrative study 2.3.

Illustrative study 2.3. Mayr & Escudero (2010)

Overview: This study examined whether learners follow individual paths in L2 vowel perception and whether the L2LP model can explain such interlearner variation.

Languages: German (TL); English (L1)

Dependent variables: (1) *Experiment 1*: Cross-language perceptual assimilation patterns; (2) *Experiment 2*: L2 vowel identification.

Independent variable: Experience with the German language as measured by their year in an academic programme (see *Participants* below)

Research questions: (1) Do L2 German learners follow different paths in their perceptual development of six rounded German vowels?; (2) Can the observed patterns be explained on the basis of the L2LP model?

Hypotheses: (1) Learners with the same L1 will vary systematically in their cross-language perception patterns; (2) Cross-language perception patterns determine L2 development as proposed within the L2LP model; (3) Similar L2 contrasts are easier to acquire than new L2 contrasts given that, according to the L2LP model, similar contrasts only require category boundary adjustments, while new contrasts require the creation of new categories.

Methodology

Participants: 15 female British English students enrolled in a German undergraduate degree. Seven were first-year students (NE1) and eight were fourth-year students (NE2).

Stimuli: (1) *Experiment 1:* listeners heard multiple tokens of 14 German monophthongs and two tokens of 13 English vowels; (2) *Experiment 2*: the German vowels alone were presented. All vowels were produced in the /bVt/ context.

Tasks: (1) Cross-language perceptual assimilation: listeners classified native and non-native vowels by selecting one of 14 words containing English vowels; (2) L2 vowel identification: listeners classified German vowels by selecting one of 14 words containing English or Germans vowels.

Main findings

Only results for the six rounded German vowels /uː, ʊ, yː, ʏ, ø, œ/ were analyzed for this study.

(1) As predicted, there was a large degree of variation across learners, as shown by the results of Experiment 1. Learners' patterns varied systematically, as predicted by the L2LP model, and were broadly characterized by containing largely either single-category assimilations or multiple neutralizations.
(2) For the majority of listeners, their cross-language mapping pattern (Experiment 1) predicted their L2 identification performance (Experiment 2), which corroborates the L2LP proposal.
(3) The more listeners assigned a German contrast to two different L1 categories, the easier it was for them to achieve native-like performance in the L2 perception of such a contrast, confirming the L2LP hypothesis that L2 learners find similar contrasts less difficult than new contrasts.

Although Mayr & Escudero (2010) suggest that the different patterns of cross-linguistic and L2 performance that they document are related to the learners' specific L1 variety, they do not provide acoustic analysis of the learners' L1 productions to support this L2LP claim. A more recent study (Escudero, Simon & Mitterer, 2012) showed that L1 variety – either North Holland or Flemish Dutch – determines both the cross-linguistic categorization of English vowels and the learners' L2 perception of English vowels. Both groups of listeners were first asked to classify TL vowels in terms of their L1 categories and were later asked to classify the same German vowels in terms of TL categories. The finding that the specific acoustic values of the learners' native variety successfully predicted their performance in the first task has been contested by Strange and colleagues, as it has been shown in many studies that comparisons between L1 and TL acoustic properties do not predict how English listeners classify French or German vowels (Strange, Bohn, Trent & Nishi, 2004; Strange, Bohn, Nishi & Trent, 2005). Indeed, Strange (2007) claims that the discrepancy between acoustic and perceptual similarity in her studies suggests that direct measures of perceptual similarity, such as perceptual assimilation, are more accurate. However, several recent studies (Gilichinskaya & Strange, 2010; Escudero & Vasiliev, 2011; Escudero, Simon & Mitterer, 2012; Escudero, Sisinni & Grimaldi, 2014) have shown that, when context and methodologies are kept constant across studies, detailed acoustic comparisons of L1 and TL categories can accurately predict cross-linguistic similarity and, therefore, can be used to predict specific developmental paths for L2 learners. These studies thus support the L2LP's hypothesis that acoustic values are predictive of cross-linguistic perceptual patterns.

To illustrate how the predictions of the SLM, PAM, and L2LP differ, consider the case of the perception of French or German /y/-/u/ by English listeners. According to the core SLM, /y/ will be perceived as a new vowel because its acoustic properties do not match English /i/ or /u/ and therefore it will not be equated to either of these L1 categories. Therefore, L2 learners will more accurately perceive and produce French /y/ than /u/, because the properties of the latter may lead them to equate it with English /u/. According to the PAM, the ability to discriminate French /y/-/u/ will depend on whether this contrast exhibits SC or CG assimilation, for which a perception experiment needs to be conducted. Finally, the L2LP would make predictions specific to a single learner based on the acoustic values of the vowels in his or her specific variety and how they compare to those of the vowel tokens of the specific variety of French or German that the learner is exposed to. For instance, the L2LP will make different predictions for learners having American, Canadian, or Australian English as an L1, as acoustic descriptions have shown that English /u/ has different degrees of fronting in these varieties (compare, for example, the values reported in Hillenbrand, Getty, Clark & Wheeler (1995) for American English and those reported in Cox (2006) for Australian English).

Having described the models' formalization of CLI and their CLI-based predictions for levels of relative difficulty, we will describe in the next section the mechanisms that each model assumes are at play during L2 learning. The resulting predicted developmental sequences of each model incorporate their notion of what scenarios lead to the greatest difficulty.

2.2.2 Learning mechanisms and developmental sequences

Within Flege's SLM, adult L2 learners retain the ability to acquire the TL sound system, as the learning mechanisms involved during native L1 learning remain intact across the lifespan (Flege, 2002, p. 224). According to the SLM, learners acquire native-like TL proficiency by developing long-term memory representations for TL categories similar to those that monolinguals develop during L1 acquisition. Despite the possibility of establishing new L2 categories, native-like proficiency is not guaranteed. The main source of the observed difficulty in accurately perceiving and producing TL categories is that L2 learners have a **common phonological space**, in which both L1 and TL categories are represented, inevitably leading to mutual influence between L1 and TL categories (Flege, 1995; 2002; 2003).

This common phonological space regularly leads to category assimilation, which is the result of equating a TL to an L1 category. In this case, TL category formation is blocked and the learner ends up with a **composite** or **merged L1–TL category**, which is defined as a long-term representation that encompasses the properties of the L1 and TL categories. This merged category is then used to perceive and produce L1 and TL exemplars (Flege, 2002; 2003). This does not mean that L2 learning will cease to occur, as L2 learners continue to be sensitive to phonetic differences between TL and L1 categories, but learning or adjustment of the merged category will inevitably affect perception and production of both languages. Two studies support the existence of a single VOT category for Italian learners of English, where properties of the two languages are merged. In these studies, learners' VOT values for English and Italian voiced (/b,d,g/; MacKay, Flege, Piske & Schirru, 2001), and voiceless consonants (/p, t, k/; Flege, 1987b) were different from those produced by monolinguals of either language. Consequently, the SLM predicts that learners who have a merged L1 and TL category will struggle to achieve native-like performance when category assimilation has taken place and will likely perform differently from monolinguals of either language.

Target-like category formation thus seems to be the recipe for success, but as Flege and colleagues demonstrate it tends to occur more often with learners whose first contact with the TL occurs early in life (before the age of 4). In §2.2.3, we will see that, with early learners, the relative amount of L1 use further predicts their L2 success. Flege (1991) found that early learners of English with Spanish as an L1 and who were living in the US produced English voiceless stops with VOT values that matched those produced by age-matched English monolinguals. However, forming a new category does not guarantee monolingual-like performance, as early learners of English with Spanish as their L1 who were exposed to English in Puerto Rico produced English VOT values that were shorter than those of English monolinguals (Flege & Eefting, 1987). In another study, the authors demonstrated that the same learners had indeed developed a new long-lag VOT for English /p/ (Flege & Eefting, 1988). Specifically, unlike English and Spanish monolinguals who produced their corresponding native-like VOT values for /p/-/b/ (long-lag versus short-lag and short-lag versus prevoiced respectively; see §5.2.1 for further discussion), the early learners produced all three different VOT categories. Importantly, it was also shown that these learners' Spanish VOT productions for /p,t,k/ were shorter than those of Spanish monolinguals, which the authors interpret as **dissimilation** – the mechanism by which L2 learners keep L1 and newly learned categories more distinct. This provides further evidence for the SLM claim that proficient bilinguals who have learned their L2 early in life will also differ from monolinguals of either language, leading to the conclusion that L2 learners will never end up with monolingual-like proficiency.

The PAM-L2 and the SLM are compatible as concerns their proposal of a common space for L1 and L2 categories (Best & Tyler, 2007, p. 26). However, the PAM-L2 argues that L2 learning occurs at two levels of extraction of articulatory gestural properties, namely the lower-level phonetic and the higher-order phonological; this has implications for L2 developmental sequences and difficulty. Specifically, L2 learners can have two phonetic realizations for their common L1-TL phonological category or phoneme, which will likely result in differences between L2 learners' phonetic categories and those of monolingual speakers. For example, an English learner of French will have the same phonological category /p/ but two phonetic realizations, namely [pʰ] for English and [p] for French. However, it is predicted that neither of these realizations will match the phonetic realization of /p/ by monolinguals of either language. This is because a common phonological space will lead to shifts in the phonetic details of the L1 and L2 versions of the common interlanguage phoneme.

On the basis of this proposal for L2 learning at the phonetic and phonological levels, Best & Tyler (2007) provide extensions to the PAM perceptual assimilation types in order to predict L2 perceptual learning. Here we discuss three of the scenarios that are comparable to those described within the SLM. In the first scenario, the PAM-L2 predicts that, when only one member of a TL phonological contrast is assimilated to an L1 category, no further phonetic learning is likely to occur. That is, the common L1-TL category can in principle shift to better accommodate both L1 and TL phonetic detail, but as the TL category will be considered a good exemplar of the L1 category, no large shifts at the phonetic level are likely to occur. In contrast, in the second scenario, if one of the members of a TL contrast is phonologically equivalent to an L1 category but has a very different phonetic realization, such as the French rhotic [ʁ] for English-speaking learners, it is predicted that L2 learners should easily form a new TL phonetic category. That is, they will notice the phonetic difference between French [ʁ] and English [ɹ] and will likely form a new phonetic category for the French realization within their common phonological category /r/. In a category-goodness scenario where one TL phoneme is a better example of an L1 category than the other, learners will more likely develop a new phonological category for the more deviant phoneme than for the one that is closer to the L1 category; this parallels the SLM proposal that new or divergent phones are easier to learn. Finally, for cases where the two phonemes in a TL contrast are equally good or bad examples of the same L1 category (single-category assimilation), the forecast is sombre, as it is likely that both phonemes will be phonetically and phonologically assimilated to the same L1 category, leading to their status as homophones.[2] The PAM-L2 proposes that the lexicon can exert a positive role in this case, as learners can form new L2 categories by detecting the contrasting function of minimally different words. Thus, it is predicted that TL phonemes that have been assimilated to a single L1 category can be discriminated if they are part of a phonologically dense neighbourhood involving many minimal pairs. That is, a growing L2 lexicon containing a large number of minimal pairs may exert a positive effect on L2 perceptual learning, allowing for the establishment of new phonological categories due to an increased pressure for discriminating words that are communicatively meaningful. Evidence for the role of the L2 lexicon on perception is provided by Bundgaard-Nielsen, Best & Tyler (2011a) who showed that Japanese-speaking learners of English with larger L2 vocabularies were more consistent in their perceptual assimilation of English vowels to Japanese vowels than learners with smaller L2 vocabularies.

Unlike the SLM and PAM that make predictions on the basis of phonetic and phonological categories, the L2LP makes a principled distinction between perceptual mappings, which result from parsing the auditory input with the perception grammar, and the abstract and discrete representations of sound categories. The L2LP model includes two levels of abstract representations, namely phonological and lexical, and two grammars that (1) map the acoustic signal to phonological representations (perception grammar) and (2) map phonological categories onto lexical representations (recognition grammar). With respect to L2 perception, it is claimed that analyzing mappings and categories separately results in an adequate description and explanation of L2 learners' difficulty and development. L2 perceptual development is accomplished through resolving two learning problems: a perceptual problem that involves changing or adjusting the L1 perception grammar so that it more successfully handles the TL input, and a representational problem, involving the creation of TL categories when needed (Escudero, 2009). According to the L2LP, when learning similar sounds, learners can *reuse* their L1 lexical categories without needing to create new ones, but will need to adjust their perception grammar. Escudero (2009) reports a case of learning similar TL categories for Canadian English-speaking learners of Canadian French, as these languages share

[2] As used in this model, this term normally refers to words but may also designate phones.

the contrast /æ/-/ɛ/.[3] It is demonstrated that advanced L2 learners are able to adjust their L1 perception grammar to match that of the TL.

Perhaps the most important difference between the L2LP and the PAM and the SLM models is that the L2LP proposes that L2 learners have separate perception grammars for their L1 and their TL. This proposal is formalized with the linguistic hypothesis of **Full Copying/Full Access**, which is advanced as the speech perception equivalent of the Full Transfer/Full Access hypothesis[4] proposed by Schwartz & Sprouse (1996) in the domain of L2 syntax and morphology. It is argued that Full Copying is the mechanism that both underlies perceptual assimilation and equivalence classification and allows learners to arrive at two distinct systems for the perception of their L1 and TL. Additionally, within the L2LP, Full Access is formalized and operationalized through the **Gradual Learning Algorithm**, which is a learning device capable of adjusting perception grammars on the basis of input (Boersma & Hayes, 2001; Escudero & Boersma, 2004; Boersma & Escudero, 2008).

In support of the proposal of separate TL and L1 perception grammars and the workings of Full Access as a learning device, Escudero & Boersma (2002) show that although advanced Dutch-speaking learners of Spanish have better TL perception of Spanish /i/-/e/ than beginner or intermediate learners, their L1 Dutch perception of the same vowels was unaffected by their TL proficiency. It was also demonstrated that, in a task where the listeners knew that the vowels were Spanish, their L1 Dutch perception varied with TL proficiency. Specifically, intermediate and advanced learners used the Dutch category /ɪ/ less often than beginner learners. The authors argue that Dutch learners adjust their Spanish perception grammar so that it no longer maps incoming input to the category /ɪ/, resulting in TL **category reduction**, while their Dutch perception grammar continues to map relevant incoming input to their Dutch category /ɪ/. Similarly, Escudero (2005; 2009) shows that, although TL proficiency affected Canadian English-speaking learners' classification of French vowels in a French task, their L1 categorization of the same vowels was unaffected by their TL proficiency when presented with an English task. This further indicates that incoming auditory stimuli are perceived differently depending on the language of the task. This evidence suggests that L2 learners can achieve native-like performance in their TL without compromising their already optimal L1 perception grammar.

3 Picard (1987) argues that Canadian French has a more fronted central vowel than European French and that, therefore, the IPA symbol /æ/ describes it better than that used for European French, namely /a/.

4 Schwartz & Sprouse propose that the L2 initial state consists of a copy of the L1 end-state morphosyntactic grammar and that learners continue to have full access to Universal Grammar; this allows for the possibility of native-like attainment under the right conditions.

With respect to cases where learners need to create a new category, the L2LP proposes that this can be done rather quickly and successfully if the TL category has acoustic differences along a dimension that has not been used in the learners' L1, as is the case with vowel duration for learners of English whose native language does not have phonemic vowel length contrasts. Escudero & Boersma (2004) demonstrate the workings of Full Access in Spanish-speaking learners who were able to distinguish the Southern British English vowels /i/ and /ɪ/ based on their duration differences and thus formed two new categories for them, namely /short/ and /long/.

In line with the proposals of the SLM and PAM that the ability to learn TL categories continues into adulthood, and based on the proposal of Full Access, the L2LP provides a specific learning mechanism that allows L2 learners to form categories, namely **distributional learning**, which is the same general mechanism that has been posited to underlie infants' phonetic category learning (Maye, Werker & Gerken, 2002; Maye, Weiss & Aslin, 2008). According to Maye and colleagues, this mechanism allows infants to harness relative frequency distributions in the continuous auditory signal when learning to discriminate speech sounds and this can be done after only two minutes of exposure to distributions of a non-native phoneme contrast. Recent neurophysiological evidence shows that adult listeners can also make use of the same mechanism rather quickly, as Spanish listeners show sensitivity to vowel duration differences in non-native vowels within a one-hour session (Chládková, Escudero & Lipski, 2013). Recent studies have also shown that adult L2 learners' perception of a new TL contrast can improve after two minutes of distributional training (Escudero, Benders & Wanrooij, 2011), and that the results of this distributional learning can be maintained for a year (Escudero & Williams, 2014). This suggests that L2 learners are capable of adjusting their perception grammars even in the most difficult case of a new TL contrast, provided that the input resembles the type of rich frequency distributions available to L1 learners. This is in line with the SLM's proposal that L2 input might be one of the major sources of variability in L2 performance, as will be discussed in the next section.

2.2.3 Variability in L2 perception

As mentioned in Chapter 1 and as you are likely aware through your own experience as an L2 learner or someone who regularly interacts with non-native speakers, it is clear that there is great interlearner variability – not all learners have the same difficulty perceiving and producing TL categories. In the previous two sections, we have discussed explanations for L2 learners' variability that stem from CLI and the specific learning scenarios that emerge based on how L1 and TL categories compare. An extra factor that was mentioned in the context of the L2LP model was dialectal differences in the perception and production of L1 and TL categories, which, as we saw above, can lead to individual differences in L2 perceptual difficulty. As mentioned in Chapter 1, these factors relating to differences between and within the TL and L1 can be referred to as linguistic factors. However, other factors outside the language domain may be at play in L2 learning. A prominent extralinguistic factor that seems to account for interlearner variation is the learners' age at the onset of acquisition (AOA). In Chapter 1, we saw that a number of researchers have proposed that there is a so-called critical period for language acquisition, leading to the suggestion that L2 learners are unlikely to attain native-like proficiency if their AOA is post-puberty.

However, Flege and colleagues postulate that the ability to learn L2 speech remains intact and that other factors confounded with AOA need to be taken into account. Although in their studies learners with a lower AOA tend to outperform those with higher AOAs, early learners do not necessarily demonstrate native-like performance. Specifically, Piske, MacKay & Flege (2001) showed that, among their group of early learners of English, those who often used their L1 Italian had significantly stronger accents when reading English sentences than those who used their L1 seldom, as measured by native-speaker ratings. These authors also report that in both these two groups, there were individuals who were scored within the native English speaker range. This suggests that amount of L1 use and not AOA predicted variability in this learner group. Flege (2003) suggests that **language dominance** may be an important factor to take into account, as Flege, Mackay & Piske (2002) reanalyzed Piske et al.'s data finding that only Italian-dominant early learners, but not those who were dominant in English, differed from native English speakers. Flege (2003) proposes that, if these results for degree of foreign accent in sentence reading also apply to TL category perception and production, L2 speech may not be irreversibly constrained, and learners who are dominant in their TL may be able to attain native-like performance.

More recently, Flege (2009) has argued that we should turn our eyes to the input that L2 learners receive, as this factor may be the key to understanding the difference between native speakers and L2 learners. For instance, Flege & Liu (2001) showed that L2 learners' type of L2 input rather than their amount of L2 use or their length of residence (LOR) in the US predicted their performance. In this study, Chinese-speaking learners of English were tested on their identification of word final consonants. Half of the learners were students, the other half had occupations that, according to the authors, reduced their interactions with native English speakers. It was shown that students with higher LOR had more native-like performances, while LOR did not make a difference among the non-student group. Although the authors did not measure the amount of L2 input each group received, both groups reported comparable L2 use, indicating that only learners who were exposed to substantial native-like L2 input (the student group) showed improvement towards native-like performance. Flege (2009) further suggests that the difficulty in estimating the exact amount of L2 input learners receive has resulted in researchers' underestimation of its predictive power. He then suggests a novel methodology for measuring L2 input in a large sample of L2 learners, including multiple measurements every day for a month using three or four questions related to learners' TL input. Although the total recording time would not amount to more than three minutes per participant per day, it is acknowledged that this type of procedure, which would allow for accurate and valid data concerning L2 input, requires large amounts of creativity, time, and financial support. This led Flege to conclude that researchers would first need to recognize the powerful predictive power of the input to L2 speech before allocating resources to its investigation.

Luckily, there are other less costly ways of testing whether L2 learners can perform like native speakers given similar types of input, as demonstrated by the recent findings within the L2LP model, indicating that L2 learners can exploit the same learning mechanism as L1 learners to form phonetic categories. Specifically, it has been shown that L2 learners are capable of extracting properties of the speech signal in order to discriminate two phonetic categories, provided that the information in the signal is as rich as that encountered by infants learning their native language. Specifically, Escudero, Benders & Wanrooij (2011) have shown that two-minute exposure to an auditory continuum featuring more frequently occurring exemplars at the extreme endpoints and less frequently occurring exemplars in the middle of the continuum led to improvement in the discrimination of a new TL contrast (one that was shown to be particularly difficult for Spanish-speaking learners of Dutch). Importantly, it was shown that improvement occurred only if the acoustic distance between the edges of the continua was enlarged or exaggerated, so that their acoustic values were more separated that those found in the natural productions of the speech sounds. This enlargement of the auditory distinction between speech sounds mimics the type of contrastive distinctions found in infant-directed speech where differences between vowels are exaggerated (Kuhl, Andruski, Chistovich, Chistovich, Kozhevnikova, Ryskina, Stolyarova, Sundberg & Lacerda, 1997).

Escudero & Williams (2014) have recently shown that the improvement gained after this very short exposure to rich vowel distributions was maintained for a year, suggesting that this mechanism is not only effective in a single laboratory session but has long-lasting effects. Importantly, it seems that not all L2 learners benefit from exposure to this rich input; Wanrooij, Escudero & Raijmakers (2013) have shown that there are very clear differences in the extent to which L2 learners can benefit from distributional learning. Importantly, the authors show very systematic differences that depend on learners' sensitivity to the acoustic properties of the TL categories, which enable the grouping of listeners according to their listening strategies and their corresponding levels of improvement. The individual strategies are also interpreted as potential developmental stages in L2 perception, which are compatible with those predicted by the L2LP model. Thus, giving the input a chance and utilizing it systematically to predict individual L2 performance is in line with Flege's most recent view on ultimate attainment for L2 speech and is at the core of the L2LP model.

In Chapters 4 through 8, we will discuss another factor that affects most L2 learners, namely their literacy skills. Unlike L1 learners, most adult L2 learners can read and write at the time they first encounter TL phonemes. As mentioned in Chapter 1, a number of studies have demonstrated the effects of native orthography on L2 speech, which is at play even when orthographic information is not available, as first demonstrated by Weber & Cutler (2004) and confirmed by Escudero, Hayes-Harb & Mitterer (2008). L2 input is not only different in richness and amount but it is both auditory and visual. The written component of the L2 input can potentially aid L2 learning. However, in many cases, orthography is associated with persistent CLI effects that may have a negative influence on perception, production, and word recognition (see Cutler (2015) for a review), leading to many individual differences among L2 learners, including different levels of non-native-like performance.

2.2.4 *Summary of theoretical concepts and models relevant to L2 perception*

CLI is at the core of L2 perception models' predictions regarding the levels of difficulty that a given L1-based learner group will have with specific TL vowels or consonants. These models differ in their proposals concerning the type of representations or invariants that L2 listeners need to master and the possibility of native-like achievement. While the SLM proposes that similar TL categories will be the most difficult to achieve, the PAM and the L2LP contend that TL contrasts that are assimilated to two L1 categories or to similar L1 contrasts will be the easiest. Unlike the SLM and PAM that concentrate on groups of highly advanced or naïve listeners respectively, the L2LP explicitly explains L2 perception at the level of the individual learner, from first contact with the TL to end state.

Regarding ultimate attainment, both the SLM and the PAM-L2 propose that L2 learners will never achieve monolingual-like performance because L1 and L2 categories co-exist in the same phonological space and, therefore, the changes that allow for more native-like L2 categories will inevitably affect the learners' L1 categories. In contrast, the L2LP proposes that native-like L1 and L2 perception is attainable, as it is demonstrated that learners can eliminate categories or shift the location of perceptual boundaries in order to achieve higher levels of L2 performance without a negative effect on their L1 perception grammar.

Finally, we discussed some of the extralinguistic factors that lead to variability in L2 learners' performance. We mentioned that the SLM considers AOA, language dominance, and LOR as potential influences but that the most important factor may be the input, as more recently suggested in Flege (2009) and as shown by studies conducted within the L2LP model. We reviewed some of the recent studies that have demonstrated that adult learners' L2 perception can improve after very short exposure to TL input that mimics the rich input that young children receive during L1 acquisition. The last factor that we discussed was the availability of written input, which can have positive or negative effects on L2 perception and production. In the next section, we will discuss models and frameworks that have attempted to explain L2 production.

2.3 Theory and L2 speech production

In contrast to the various L2 perception theories discussed in the previous sections, there is no comprehensive theory of L2 phonological production. This is not to say that work on production has not been theoretically informed. Rather, as opposed to developing a unified model, researchers have focused on various aspects of acquiring L2 speech production and, for each of these, have drawn on theories and principles from phonetics, phonology, and L2 acquisition theory more generally. In what follows, we will review the various theoretical models and principles that have been used most often in L2 speech research. As we did for perception, our discussion will be structured vis-à-vis some of the main research themes that we discussed in Chapter 1.

2.3.1 Cross-linguistic influence: Articulatory settings theory

The most regularly investigated phenomenon in L2 speech production is cross-linguistic influence, the effect of previously acquired languages (the L1 or one or more L2s) on learners' realization of segmental and prosodic phenomena. In many cases, researchers simply assume or predict that CLI will shape learners' production without discussing the specific mechanisms that underlie this influence, for example, learners' dependence on L1 articulatory patterns or phonological representations. While such predictions are grounded in empirical fact, they lack explanatory adequacy: they do not explain what is the source of such influence. As we will discuss here, there is, however, at least one theory that has been used in L2 speech research to formalize CLI in a way that explains why a speaker's other language(s) shape(s) L2/L3 production.

Articulatory Settings theory (AS; Honikman, 1964)[5] proposes that every language is characterized by a series of general articulatory positions involving the lips, tongue, cheeks, jaw, and pharynx. Honikman defines an articulatory setting as "the overall arrangement and manoeuvring of the speech organs necessary for the facile accomplishment of natural utterance... [that] determines the phonetic character and specific timbre of a language" (p. 73). According to this author, a language's articulatory setting, which involves sub-settings for articulators, both external (i.e., lips) and internal (all articulators within the oral cavity), is shaped to a great extent by the most common consonants and vowels as well as by its phonotactics. In English, it is thus coronal consonants – first and foremost [t,n] followed by [s] then [d,ɾ,z] – that determine the basis of the internal articulatory setting. Among the internal organs, the tongue is most important, whose own setting involves a point of anchorage – for example, a point on the roof or floor of the mouth such as the upper posterior pre-molars or lower front teeth respectively – and a free, operative part such as the tip or blade. A language's articulatory setting is also characterized by the parameters of muscular tension of the tongue, lips, cheeks, jaw, and pharynx; the degree of pressure between the active and passive articulators in consonants with median closure (i.e., stops, affricates, nasals, and laterals); and general jaw position (p. 77). Languages may or may not share their articulatory settings. For example, Honikman characterizes overall lip rounding as 'energetic' and 'vigorous' in both French and German versus 'neutral' and 'unvigorous' in English and 'close-spread' in Russian. She underlines the importance of articulatory settings for L2 speech learning: when the L1 and TL differ, learners must acquire the new articulatory settings of the TL in order to effectively coordinate the continuum of articulations necessary for fluid speech in native pronunciation. Indeed, given the overall importance of a language's articulatory setting, she proposes that L2 speech learning may be facilitated by focusing first on articulatory settings, followed by the specifics of individual segments (p. 80).

While the concept of AS has received much theoretical consideration including in applied linguistic studies on corrective phonetics and L2 pronunciation teaching (e.g., Wenk, 1979; Jenner, 1992; Mompeán González, 2003) and has been tested indirectly via acoustic studies (e.g., Lowie & Bultena, 2007 for Dutch-speaking learners' English vowels), relatively few studies have investigated it directly via articulatory studies. Gick, Wilson, Koch & Cook (2004) tested the hypothesis that language-specific articulatory settings exist based on an X-ray film comparison of five Canadian English and five Québécois French native speakers' reading of

[5] See Jenner (2001) for a discussion of the history of the concept. One encounters various related terms including the long-established concept of **base of articulation** as well as **phonetic settings** (e.g., Laver, 1994), and **voice quality settings** (e.g., Esling & Wong, 1983; Laver, 2009). See Borrissoff (2011) for an extensive discussion of the similarities and differences between these various concepts.

individual words and sentences. In the first of two experiments,[6] the researchers measured upper and lower lip, pharynx, velum, tongue, and jaw positions during interutterance rest position, which allows for the distinguishing of a language's AS from the positions required for particular segments or sequences. With the exception of velopharyngeal port width and jaw aperture, they found significant between-language differences. Interestingly, the authors highlight that their measurements are in keeping with the majority of differences Honikman (1964) described between English and French. They conclude by stating that their findings are consistent with the existence of language-specific articulatory settings that affect language acquisition, among other phenomena.

More directly related to L2 speech learning, Wilson & Gick (2014) investigated AS in eight self-identified bilinguals. Their spoken English and French proficiency was evaluated by from seven to ten native speaker judges of each language on a five-point Likert scale with two groups being identified: Group 1 included four participants who were rated as native speakers of both languages by at least two judges; Group 2 involved four individuals identified as native speakers in only one of the languages (n = 3) or neither (n = 1). The researchers obtained articulatory measurements of the tongue using ultrasound and of the lips and jaw via optical tracking in interspeech posture (i.e., the active position of the articulators between utterances). Each participant read a block of 30 sentences twice in each language (monolingual mode) then a mixed block containing 15 sentences in each language (bilingual mode). Based on the measurement of two parameters (tongue tip height, lower lip protrusion) found to distinguish English and French in Wilson's (2013) parallel study of monolinguals using the same methodology, Wilson & Gick report two general findings. First, in monolingual mode, for the four participants who were rated as native speakers of both English and French, there were significant between-language differences in both parameters (two speakers) or lower lip protrusion alone (two speakers). These findings paralleled differences found for monolingual speakers of both languages in Wilson (2013). In contrast, the four participants who were not judged to be native speakers of both languages did not pattern with Wilson's (2013) speakers. Second, in bilingual mode, for three of the bilingual group speakers, the interspeech posture resembled that of the monolingual mode of their most-used language. For the most balanced bilingual, there was a hybrid of French lip and English tongue-tip inter-speech postures. In summary, the more bilingual a speaker, the more distinct their articulatory settings for each of the languages spoken.

[6] In the second experiment, they measured the same articulatory parameters with the exception of lip rounding during the articulation of /i/ and compared these to those of rest position. This experiment tested another principle that is not directly relevant here.

AS provides an explanatory framework for understanding the physical mechanisms that underlie CLI. Given the increasing availability of articulatory imaging techniques such as ultrasound (see Gick, Bernhardt, Bacsfalvi & Wilson, 2008 for an introduction) and electromagnetic (midsaggital) articulography (EMA/EMMA; see van Hoole & Nguyen, 1999 for an overview), such work is becoming more and more possible (see Mennen, Scobbie, de Leeuw, Schaeffler & Schaeffler, 2010 for an overview of these methods and their potential use for measuring phonetic settings; see Ramanarayanan, Goldstein, Byrd & Narayanan, 2013 for the measurement of AS using magnetic resonance imaging (MRI)).

Illustrative study 2.4. Święciński (2013)

Overview: This study investigated whether there are articulatory setting (AS) differences between English and Polish, and whether native Polish speakers who acquire the English AS have less pronounced foreign accents.

Languages: English (TL); Polish (L1)

Dependent variable: Perceived English foreign-accentedness – participants were subjectively classified into two groups by the researcher as having 'heavily accented' versus 'near-native' pronunciation

Independent variable: English articulatory settings (see *Measurements* below)

Research question: Do between-speaker differences in English articulatory settings correlate with differences in perceived English foreign accent?

Hypotheses: This was a descriptive study with no hypotheses.

Methodology

Participants: 4 adult native speakers of Polish (2 male, 2 female) who had studied English for a minimum of 8 years and whose English proficiency ranged from B2 (Upper intermediate) to C2 (Mastery) on the scale of the Common European Framework of Reference for Languages. In terms of perceived English accentedness, 2 were classified as 'heavily accented', 2 as 'near native'.

Task: Participants had to provide answers (most often a single word) to 10 general knowledge questions (e.g., *What is the capital of Spain? How much is 2 plus 8?*) presented aurally every 6 seconds in both Polish and English.

Measurements: Using three-dimensional electromagnetic articulography that involves sensors secured to the articulators with a medical adhesive, measurements of the lower and upper lips, the jaw, and the tongue (tip, front, and back) were taken every 5 ms during the speech-ready posture (SRP) that immediately preceded learners' responses to the questions.

Data analysis: Horizontal and vertical movements measured in millimetres were calculated for each sensor/articulator. Mean positional values of the sensors were then calculated for each speaker by language.

Main findings

(1) For the native Polish speakers whose English was classified as near native, statistically significant differences existed for tongue position of their Polish and English SRPs (Speaker 1: greater fronting of the central tongue line and greater raising of the tongue front and back in English; Speaker 2: greater tongue raising in English).
(2) No such differences existed for those learners whose English was characterized as highly accented.
(3) No differences in lip and jaw positions were found for any of the 4 English learners. In general, there appeared to be a positive correlation between differences in L1-TL articulatory settings and perceived foreign accent.

2.3.2 Developmental sequences and relative difficulty of acquisition: Articulatory constraints and markedness

As was discussed in §1.3.3, L2 learners, like their L1 counterparts, most often do not acquire new segments and prosodic phenomena in an all-or-nothing manner. Rather, certain structures are acquired first and there may be a consistent developmental order observed across learners, even those who do not share an L1. As we will see in the following two sections, studies that have examined developmental patterns or that have sought to determine why certain phonetic and phonological structures are acquired with relatively greater ease have drawn on both phonetic and phonological principles.

2.3.2.1 Articulatory constraints

Phonetic research has demonstrated time and time again that certain articulatory combinations and speech sounds or sequences are more easily realized and thus are more common cross-linguistically. For example, stop consonants are more likely to be voiceless, front vowels unrounded, and syllables vowel-final. Moreover, some tendencies may be contextually conditioned. For example, voiced stops, when they occur in a language, are more common syllable-initially than syllable-finally. Each of these tendencies can be explained with reference to phonetic production constraints.[7] In the case of stop consonant voicing, there is an articulatory conflict between the segment's manner and voicing: oral occlusion of the airstream makes it difficult to maintain the oral-subglottal air pressure difference necessary for voicing. This is particularly true when the stop is followed by another stop or

[7] Human speech perception may also shape such tendencies. For example, cues to consonant place of articulation include formant transitions with the preceding and following vowels (see §5.2.1). Place of articulation is thus more difficult to perceive in syllable-final consonants – whether pre-consonantal or pre-pausal – which lack a following vowel. See Diehl & Lindblom (2000) for general discussion.

a pause (i.e., syllable-finally). The preference for unrounded front vowels is the result of conflicts between two articulations: rounding of the lips naturally involves backing of the lower jaw and consequently of the tongue (dorsum), whereas anterior vowels require a relatively more fronted tongue position.

Phonetic constraints shape L2 learners' production both in terms of developmental sequences and relative accuracy (e.g., Yavas, 1997; Colantoni & Steele, 2007a; 2008). For example, L2 learners whose L1 does not allow syllable-final stops will first acquire voiceless stops before their voiced counterparts and – even when learners can realize both/all members of a contrast they may be more accurate with one member (e.g., more target-like realizations of voiceless /p,t,k/ than /b,d,g/ syllable-finally; see Broselow (2004) for discussion of relevant studies). Given the vast range of phonetic constraints that shape human speech production, it is not possible to list them all here. What is recommended is that, once you have determined the phenomenon that you wish to study, you read articulatory studies on the phenomenon as well as general principles and theories of speech production (e.g., Lindblom, 1989; Ohala, 1993) in order to become aware of the relevant constraints.

Illustrative study 2.5. Yavas (1997)

Overview: This study examined devoicing patterns in L2 learners' production of syllable-final English voiced stops /b,d,g/. Specifically, the researcher investigated the contribution of stop place of articulation (POA) and height of the preceding vowel to variability.

Languages: English (TL); Japanese, Mandarin, Portuguese (L1s)

Dependent variable: % stop voicing calculated by dividing the duration of voicing by the total duration of the stop

Independent variables: *Linguistic*: (1) Stop POA [bilabial, alveolar, velar]; (2) Height of preceding vowel [high, non-high]

Hypotheses: The % voicing of the learners' stops will (1) decrease the further back the POA of the consonant (i.e., bilabial > alveolar > velar) and (2) will be lower with a preceding low versus high vowel

Methodology

Participants: 19 native speakers of Japanese (n = 5), Mandarin (n = 9), and Portuguese (n = 5) as well as 4 native English speaker controls, all of whom were studying at an American university at the time of testing. The L2 learners were chosen from a larger group based on their performance on two reading tasks (word list, passage): only those whose speech production involved syllable-final devoicing were included in the main study.

Task & stimuli: All participants read a list of 12 monosyllabic common words ending in a voiced stop controlled for voicing (4 each of /b,d,g/) and the preceding vowel; within each group of 4, there were two high vowels (/ɪ/ or /ʊ/) and two non-high vowels (/æ/, /ɛ/, or /ʌ/).
Data analysis: For each of the 12 word-final voiced stops produced by the L2 learners and native speakers, stop duration and voicing were measured acoustically.

Main findings

In terms of % voicing of the word-final stops,

(1) no main effect was found for speaker language nor an interaction effect between language and POA or vowel height.
(2) For the 19 L2 learners as a group, an interaction effect was found between POA and vowel height such that (a) there were significant differences between POA with high vowels (Bilabial: 28%; Alveolar: 24%; Velar: 18%) but not with non-high vowels; (b) in alveolars and velars but not bilabials, % voicing was significantly greater with low than high vowels (Alveolars: 29% versus 24%; Velars: 27% versus 18%). For 7 of the 19 L2 learners, the reverse held for this latter asymmetry; (c) compared to the native speaker controls, the learners devoiced 50% more for each of the POA.

2.3.2.2 Markedness

Markedness is a measure of relative phonological complexity. For a set of structures in a markedness relationship, one will be **unmarked**, the other(s) relatively more **marked**. The term 'unmarked' can also be understood as 'simpler' or 'more natural'. Phonologists determine which structure is unmarked based on various criteria including implicational universals, cross-linguistic frequency, and L1 acquisition orders (see Rice, 2007 for discussion). Of these three criteria, implicational universals are normally given the greatest weighting, at least for the formalization of markedness adopted by L2 acquisition researchers. An implicational universal takes the form of 'If A, then B'. A commonly cited example is phonemic voicing in stops. Cross-linguistically, with very few exceptions, if a language has phonemically voiced stops such as /b,d,g/, then it also has phonemically voiceless stops such as /p,t,k/. This is the case in many languages including English, French, and Spanish. In contrast, the reverse situation does not hold: the presence of /p,t,k/ does not imply the presence of /b,d,g/ – languages may have phonemically voiceless stops alone. Accordingly, voiceless stops are said to be typologically unmarked compared to voiced stops. In order to establish markedness relationships, linguists rely on cross-linguistic typologies. In the case of phonology, much of the work on implicational universals is based on the work of Greenberg and colleagues (e.g., Greenberg, 1965; 1966; 1978). When the unmarked member is widely attested cross-linguistically, it is referred to as a **universal**. For example, in terms of syllable structure complexity, consonant-vowel (CV) syllables are less marked than consonant-vowel-consonant (CVC) or any other type of syllable (CCV, CVCC, etc.). Moreover, all languages permit CV syllables. The CV is thus sometimes called the 'universal syllable'.

Markedness was first integrated in L2 phonological theory into Eckman's (1977) Markedness Differential Hypothesis (MDH). While Lado's (1957) Contrastive Analysis Hypothesis (CAH) predicated that TL structures absent in a learner's L1 should be difficult whereas those having an L1 equivalent should be easy, many counterexamples were found. Eckman thus proposed revising the CAH by integrating implicational markedness as a measure of relative difficulty. The MDH proposed that (1) TL structures that both differ from the L1 and are implicationally more marked will be difficult; (2) relative difficulty equates with relative markedness; and (3) those structures that differ but are less marked will not be difficult (Eckman, 1977, p. 321).

The role of markedness in L2 phonological acquisition has been well investigated (see Eckman, 2008 for an overview; see, e.g., Broselow & Finer, 1991; Major & Faudree, 1996; Carlisle, 1998 for empirical studies) including the role of universals (e.g., Benson, 1988). Some of this research has tested the MDH directly, not always providing (complete) support for the hypothesis (e.g., Cichocki, House, Kinloch & Lister, 1999; Chan, 2007; Colantoni & Steele, 2008).[8] Another body of work has placed markedness at the core of L2 phonological learning by adopting the framework of Optimality theory (see Hancin-Bhatt, 2008 for an overview of Optimality theory and its use for modelling L2 phonological production; see, e.g., Broselow, Chen & Wang, 1998; Hancin-Bhatt, 2000; Lombardi, 2003; Broselow, 2004; Bunta & Major, 2004; Cardoso, 2011b for empirical studies).

It should be noted that there are at least two limitations to typological markedness (see Haspelmath, 2006 for a broader discussion of limits). First, in contrast to phonetic constraints that are grounded in general principles including those of aerodynamics, markedness in and of itself lacks explanatory adequacy – it does not explain why structure A is relatively unmarked compared to structure B cross-linguistically. Such explanations must be grounded in other principles including those of perceptual and articulatory phonetics (see, e.g., papers in Hayes, Kirchner & Steriade, 2004 for natural language; see Colantoni & Steele, 2008 for L2 phonology). Second, given that determining the relative markedness of two structures requires them to be in an implicational relationship, markedness cannot be

[8] Eckman (1991) subsequently forwarded the Structural Conformity Hypothesis, a weaker version that proposed simply that 'the universal generalizations that hold for the primary languages hold also for interlanguages' (p. 24).

used for structures where no such relationship exists. This limits the phenomena that can be studied using this framework. Within research on L2 phonology, markedness studies have indeed focused almost exclusively on stops and syllable structure. These are phenomena for which universals are well established (see §5.1 and §7.1 for examples).

Illustrative study 2.6. Chan (2010)

Overview: This study tested Cantonese-speaking learners on their production of two- and three-member English onsets in order to test the MDH, focusing specifically on two universals from Greenberg (1978): (1) Every initial or final sequence of length *m* contains at least one continuous subsequence of length *m*-1 (e.g., if a language allows three-member onset clusters such as /str/, it will also allow /st/, /tr/ or both); (2) In initial systems, the existence of at least one cluster consisting of obstruent+nasal implies the existence of at least one cluster consisting of obstruent+liquid.

Languages: English (TL); Cantonese (L1)

Dependent variable: % of target onset productions

Independent variables: *Linguistic*: (1) Number of consonants [2, 3]; (2) Segmental composition (i.e., type of obstruent–approximant combination)

Hypotheses: Following the MDH and using the criterion of 80% target production as the threshold for a cluster having been acquired, (1) all participants who accurately produce 80% or more of a three-member onset (e.g., /spɹ/) attempted will also produce 80% or more of at least one of its corresponding two-member subsequences (i.e., /sp/, /pɹ/); (2) all participants who accurately produce 80% or more of one of the obstruent+nasal onsets attempted (e.g., /sm/) will also accurately produce 80% or more of one of the obstruent+liquid onsets attempted (e.g., /sl/).

Methodology

Participants: 12 Cantonese-speaking learners divided into two groups:
(1) 6 intermediate-proficiency high school students (mean age 16 years);
(2) 6 advanced-proficiency undergraduate students (mean age 23 years). All had begun learning English at 4/5 years of age.

Tasks & stimuli: Four tasks were chosen to elicit different speech styles: (1) *Word list reading*: This included 212 primarily mono- and bisyllabic words including the four types of onsets relevant to testing the two universals (obstruent+nasal, e.g., *snow*, *smear*; obstruent+liquid, e.g., *climate*, *fry*; three-member and corresponding two-member onsets, e.g., *scratch*, *scar*) principally in word-initial position; (2) *Passage reading*: Participants read three passages of approximately 350 words each containing a variety of the target onset clusters; (3) *Picture description*: Participants had to describe 147 images depicting an object, action, or scene corresponding to a word beginning with a target onset cluster (e.g., *flag*, *squeeze*). This task was designed to elicit similar words to those of the word list but without the influence of orthography; (4) *Interviews*: five-minute informal interviews on personal topics (e.g., memorable life events, holidays, preferred movie star) were conducted by a research assistant.

Data preparation & analysis: The recordings of all tasks were transcribed by two individuals resulting in a data set of 7,796 tokens representing 28 different onset clusters. Where between-rater differences existed, transcriptions were reviewed by both individuals and corrected. For the 1.4% of cases where an agreement could not be reached, the tokens were excluded from further analysis. Each token was coded as a non-target or target production, then for each cluster type, the % of target productions was calculated by participant.

Main findings

(1) Overall, participants were more accurate with two- than three-member onsets on all tasks. Consistent with the first universal, in all cases where a learner had acquired a given three-member onset, s/he had also acquired at least one and often both of the two-member subsequences.

(2) Participants were overall more accurate with the two obstruent+nasal (/sn/, /sm/) than the obstruent+liquid onsets (the highest mean % of target productions (89%) was observed with the parallel /sl/ onsets). Nonetheless, consistent with the second universal, all participants had acquired both the obstruent+nasal and at least one obstruent+liquid onset.

2.3.3 The interaction between CLI, universals, and input: The Ontogeny and Phylogeny Model

L2 speech learning is influenced by a learner's L1, the TL, and certain universal processes common to language learning and human language more generally. Among many others, universal processes include syllable structure modifications such as coda devoicing (e.g., *bad* pronounced [bæt]), vowel epenthesis (e.g., *bad* pronounced [bædə̠]), and consonant deletion (e.g., *bad* pronounced [bæ]). All of these processes result in less marked structures – voiceless codas are unmarked compared to their voiced counterparts, and CV syllables are unmarked relative to CVC sequences (see §7.4 for detailed discussion of these processes in L2 speech perception and production).

Major's (2001) Ontogeny and Phylogeny Model (OPM), an extension and elaboration of his earlier Ontogeny Model (OM; 1986, 1987a), formalizes the interaction between L1-based CLI, the TL, and universal processes. As its name indicates, it models both individual (ontogeny) and group (phylogeny) behaviour in terms of how either an individual's or group's language changes over time; the latter includes historical change and dialectal variation. In terms of the ontogeny component relevant to L2 speech learning, the OPM proposes that an interlanguage is a dynamic system shaped by a learner's L1, the TL, and universals, which include not only abstract linguistic constructs such as those proposed by phonologists but also "anatomical, functional, and processing properties of the human mind" (2001, p. 83). The model has four corollaries that specify the type and rate of interaction, including the way in which they affect speech style. The Chronological and Stylistic corollaries stipulate that, over time or with increased stylistic formality (e.g., read versus spontaneous speech), L1 influence decreases, that of the TL increases, and universals increase then decrease. With TL phenomena that are similar to those in the L1 (Major provides the example of English alveolar [t] versus Spanish dental [t̪]), change is slow (Similarity Corollary). With marked phenomena, whereas L1 and TL influence is slower than what is predicted by the core Chronological Corollary, that of universals increases more rapidly followed by a slower decrease (Markedness Corollary). Overall, the model proposes dominant L1 influence at earlier stages, the emergent influence of the same universals that affect L1 acquisition and language change with time, and the possibility – even if rare – of target-like behaviour at later stages. Major stipulates that for some very marked phenomena, it is possible that little or no acquisition will occur with some learners (i.e., L1 influence, with or without the presence of universal processes, will continue to dominate). He also highlights that individual differences related to language aptitude, and psychological and social factors including the degree of speech monitoring may result in interlearner variability. Finally, Major claims that the model is valid for multilinguals for whom 'L1' can be replaced with 'L1 plus L2/L3/etc.' The OM and OPM have been used as a theoretical framework (e.g., Major, 1999; Díaz Collazos & Pascual y Cabo, 2011; García, 2013) or parallels have been drawn between research findings and the claims of these models (e.g., Hancin-Bhatt & Bhatt, 1997; Boudaoud & Cardoso, 2009).

Illustrative study 2.7. Major (1999)

Overview: This study tested the earlier version of the OPM, Major's (1986, 1987a) Ontogeny model, by investigating the acquisition of English consonant clusters.

Languages: English (TL); Brazilian Portuguese (L1)

Dependent variable: % of target-like, transfer-based, and development-based cluster realizations

Independent variables: *Linguistic*: (1) Position of the cluster in the word [initial, final]; (2) Speech style [less formal, more formal]; *Task*: (3) Time of recording

Hypotheses: Following the OM's general predictions, L1-based influence should decrease over time whereas universal developmental processes should first increase then decrease.

Methodology

Participants: 4 adult (2 female, 2 male) L1 Brazilian Portuguese speakers enrolled in an intensive (approximately 40 hours per week) beginner-level English course in their home country.

Stimuli: Words containing two-member word-initial (/s/+lateral or stop, stop-liquid; e.g., *slaps*, *spy*; *class*, *train*) or word-final (liquid-stop, obstruent-obstruent; e.g., *heart*, *bulb*; *leaks*, *tapped*) clusters; the same words were used in the same order in the word list and the text.

Tasks: Participants read the word list and text after listening to a recording of an English native speaker that served as a model to familiarize them with all words and to avoid spelling-based pronunciations. The researcher then engaged them in a conversation. The procedure was repeated three times at approximately four-week intervals.

Data analysis: Given the learners' low level of English proficiency, the conversations generated too few data points for analysis. Recordings of the two reading tasks were transcribed by the researcher and an assistant; intertranscriber agreement for the clusters ranged from 90% to 95%. Each cluster was then classified as target-like (C) or as a transfer-based (T; e.g., epenthesis of the Portuguese epenthetic vowel [i]; e.g., [i]sky or hard[i]) or developmental (D; final obstruent devoicing) substitution.

Main findings

(1) Consistent with the OM, over time, the number of C forms increased, of T forms decreased but of D forms remained essentially the same. In terms of cluster position, there were more T forms with initial clusters, more D forms with their final counterparts.

(2) There was very little difference between the word list and text and, thus, no support for the OM's claims concerning style. Major proposes that this may be due to the fact that words in the word list were preceded by pauses as opposed to other segments, which favoured more target-like production. Moreover, the formality differences between the two types of reading tasks might not have been sufficient for style-based effects to have emerged.

2.3.4 Variability: Variationist approaches

As we discussed in §1.3.4, one of the most salient characteristics of L2 acquisition is the high degree of variability observed both between learners and within the production of a single learner. As such, it is not surprising that L2 speech researchers have drawn on Variationist sociolinguistic theory that models a speaker's competence as a highly variable system conditioned by linguistic, speaker, task, and other variables (see Bayley & Tarone, 2011 for an overview). Indeed, the first L2 study adopting this framework focused on interlanguage phonology (Dickerson, 1975). In the case of phonology, linguistic variables can include segmental features (e.g., voice, POA, manner), prosodic features such as stress, and aspects of the linguistic environment (e.g., preceding and following consonants or vowels). Speaker variables include characteristics such as an individual's age, gender, and socio-economic status (see Hansen Edwards, 2008 for discussion). In many Variationist studies, variability is formalized quantitatively via the probabilistic weighting of each variable that (dis)favours the production of a given variant (see Illustrative study 2.8 below for an example). Weightings are determined by multivariate statistical analysis, often using the variable rule software Varbrul or GoldVarb (see Tagliamonte (2006) for general discussion and Young & Bayley (1996) for L2 research in particular).

Within Variationist research, L2 speech researchers have investigated two types of variability. The first involves variation in L2 learners' production that has no counterpart in the target language (e.g., simplification of consonant clusters via epenthesis and deletion). To date, phenomena studied have primarily involved consonants (e.g., Rau, Chang & Tarone, 2009) including syllable onsets and codas (e.g, Ross, 1994; Bayley, 1996; Hansen, 2001; 2006; Hansen Edwards, 2011). Some work has undertaken Variationist analyses of L2 learners' production within Optimality Theory (Boudaoud, 2008; Cardoso, 2011b). The second type of variation that has been modelled with this approach involves variable TL phenomena (e.g., Adamson & Regan, 1991; Thomas, 2002; Major, 2004; Uritescu, Mougeon, Rehner & Nadasdi, 2004; Howard, Lemée & Regan, 2006; Boudaoud & Cardoso, 2009). For example, Major (2004) investigated the influence of speaker gender and speech style on the production of four casual speech processes in American English – palatalization (e.g., *got you* → go[tʃ] you), deletion of the /v/ in *of*, the alternation between *–ing* and *–in'* (e.g., *going* versus *goin'*), and /n/ assimilation with *can* (e.g. *he ca*[ŋ] *go*). Major found that gender and speech style shaped the production of his Japanese- and Spanish-speaking L2 learners differently than that of the native English speakers. Whereas speech style had a much greater effect on variability with the latter, with the Japanese speakers its effect was negligible compared to a small but significant effect for gender, and with the Spanish speakers style and gender played an equal role. We will discuss this study further in §5.5.3 with reference to variability in the acquisition of fricatives.

As can be seen from this brief overview, Variationist analyses offer many tools for the study of L2 speech variability. One of the greatest of these is the range of independent variables that can be studied, including factors discussed in Chapter 1 such as quality and quantity of input. The quantitative nature of the model is also very appropriate for experimental phonetic research.

Illustrative study 2.8. Howard, Lemée & Regan (2006)

Overview: This study compared the acquisition of variable /l/-deletion in the French personal pronouns *il, ils* 'he/it-IMP, they (M)' and *elle, elles* 'she, they (F)' by two groups of English-speaking learners differing in study abroad experience. In native French, /l/-deletion is variable, being favoured by both linguistic variables (e.g., 3rd-person pronouns, a following consonant) and speech style (e.g., informal registers).

Languages: French (TL); English (L1)

Dependent variable: % /l/-deletion

Independent variables: Based on previous research on the phenomenon in native speech, the following factors were investigated: *Learner*: (1) Speaker gender [male, female]; *Linguistic*: (2) Following segment [vowel, consonant]; (3) Preceding segment [vowel, consonant, pause]; (4) Category of following word [verb, pronoun, other]; (5) Pronoun [*il*-personal, *il*-impersonal, *ils*, *elle*, *elles*]; (6) Position and distance of co-referent [no co-referent, following, preceding, 1 clause right, 1 clause left, 2 or more clauses away, unknown]; *Task*: (7) Style [formal, informal]

Research questions: (1) What are the characteristics of the acquisition and use of /l/-deletion by instructed L2 speakers?; (2) Does contact with native speakers have a positive effect on L2 speakers' acquisition of phonological variation and sociolinguistic competence?; (3) Do the L2 learners' constraint weightings match those of native speakers?

Methodology

Participants: 19 Irish English-speaking L2 learners in their final year of an undergraduate French programme (aged 19–21 years). All had 8–9 years of French instruction at the time of the experiment. The majority (n = 15) had spent their third academic year in France.

Task: Sociolinguistic interviews, approximately one hour in length, were conducted individually with each participant. Both informal and formal topics were discussed including visits to France, hobbies and pastimes, and employment and university studies.

Data analysis: All tokens of *il*, *ils*, *elle*, and *elles*, except those followed by an /l/-initial word in order to avoid phonetic ambiguity, were coded for all of the relevant independent variables. A step-wise regression analysis was then run using GoldVarb.

Main findings

(1) All of the factors except gender, style, and the position and distance of the co-referent were shown to condition /l/-deletion. For the study abroad group,

deletion was favoured (a) before a consonant, (2) following a vowel or pause, (3) before a pronoun, and, to a lesser degree, before *ne* '(NEG)', and (4) with impersonal *il* and, less so, personal *ils* and *elles*.

(2) The rate of /l/-deletion was greater in those speakers who had spent their third year abroad (33% versus 6%).

(3) The L2 learners' rates of deletion were vastly lower than those reported previously for native European French speakers (88–100%). Nonetheless, some of the linguistic factors that favoured /l/-deletion in previous studies of native speakers (pronoun, following segment) also favoured deletion in the L2 speaker data.

2.3.5 *Summary of theoretical concepts and models relevant to L2 speech production*

In this section, we have reviewed a number of theoretical concepts and models relevant to the study of L2 speech production. As concerns their general nature, they range in terms of whether they are based primarily in phonetics (articulatory settings and articulatory constraints) to those that are more phonological (markedness). This has consequences for the types of learning phenomena to which they can be applied. For example, whereas articulatory settings can primarily be used for the study of CLI, markedness lends itself to the study of development. Some of the models can be adapted to the study of multiple phenomena: both Major's OPM and the Variationist framework are relevant for transfer and development.

2.4 Chapter summary

This chapter began with a general discussion of the properties of scientific theories as well as how and when they are used, then turned to examining the specific characteristics of an overarching theory of L2 speech learning. In the two main sections that followed, we looked at theoretical principles, concepts, and models for studying L2 speech perception and production. In the case of perception, we saw that, whereas all theories proposed a strong role for CLI and for L1–TL comparison of similarity when predicting relative difficulty, they differed in their primitives (phonetic, phonological, or both), in the structures that they targeted (phonemes, position-specific allophones, phonemic contrasts, phonological versus lexical representations), and the ways and extent to which they elaborated how perceptual learning occurs. In the case of production, lacking a model particular to L2 speech, we discussed how research has drawn on work in phonetics (articulatory settings, constraints on consonant production), phonology (markedness), L2 acquisition (Major's Ontogeny and Phylogeny model), and Variationist sociolinguistics.

In Part II of the book (Chapter 3), we will turn to how these theoretical principles, concepts, and models can be tested empirically in studies of L2 speech perception and production. Adding this 'practical' knowledge to the more abstract and theoretical knowledge of the current chapter will be essential for understanding and evaluating the research to be presented in Chapters 4 through 8 as well as for carrying out independent research.

2.5 Keywords

formal model, adequacy (descriptive, explanatory, predictive), types of studies (descriptive, analytical), category (new, similar, merged), perceptual foreign accents, Phonological Interference Model, distinctive features, Speech Learning Model, equivalence classification, category formation, perceptual assimilation, Perceptual Assimilation Model, two-category assimilation (TC), single-category assimilation (SC), category goodness assimilation (CG), uncategorized–uncategorized (UU) contrast, categorized–uncategorized (UC), non-assimilable (NA) contrast, Second Language Perception Model, perception grammar, common phonological space, Full Copying/Full Access, Gradual Learning Algorithm, distributional learning, language dominance, articulatory settings, phonetic constraints, markedness (unmarked versus marked features/structures), universals, typology.

2.6 Review questions

(1) In general, what are the qualities of a good theory? What advantages and, potentially, disadvantages are there to using theory when conducting scientific research?
(2) What general phenomena does a theory of L2 speech research need to account for? Discuss with a specific segmental or prosodic example.
(3) If you wished to study the L2 acquisition of the tense–lax contrast in English vowels (e.g., *sit–seat*, *debt–date*), which theoretical concepts and frameworks might you use and what would be their predictions for your potential findings?
(4) Provide an example of a TL segment, prosodic phenomenon, or contrast that you (or learners that you know) have found particularly difficult to learn but that has not been mentioned in this chapter or in Chapter 1. Provide three explanations for this difficulty based on the factors that were discussed within the perception and production frameworks (e.g. CLI, markedness, AOA, etc.).
(5) In what ways are the theoretical concepts and models for L2 speech perception and production similar? In what ways do they differ?